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National Geospatial-Intelligence Agency (NGA)
(U) NATIONAL SYSTEM FOR GEOSPATIAL INTELLIGENCE
(NSG) ENTERPRISE
TEST AND EVALUATION MASTER PLAN (TEMP)
Revision D

13 June 2018

Abstract: (U) The NSG Enterprise TEMP describes the independent life cycle, enterprise-level testing of the NSG and Allied System for Geospatial-Intelligence (ASG).

Key Words: (U) National System for Geospatial-Intelligence (NSG), National Geospatial-Intelligence Agency (NGA), NGA Test Organization (NTO), NTO Test Environment, System Test, Test and Evaluation (T&E), Testing, Image Quality, Product Quality, NGA Image Quality and Utility (NIQU)

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023	18 Dec 2014	(U) Change log, Effectivity table, added requirements tables, and Table 4-1.	RFC NSG-0046JR1 MTRB 0056-14 dtd 19 Dec 2014
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024	4 Mar 2015	(U) Change Log, Table 4-1	RFC N52-0015J MTRB 0013-15 dtd 20 Mar 2015
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C/N 010		This Change Notice incorporates the following RFC:	
032	15 Jun 2017	(U) Change Log, effectivity NE509 is baselined with this RFC. Changes reflect this. Deleted the effectivity page and the effectivity tags in table 4.	RFC NBN-0001 MTRB 0023-17 dtd 14 Jun 2017
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C/N 012		This Change Notice incorporates the following RFCs:	
034	22 Mar 2018	(U) Change Log, Appendix B, Appendix D. Note: REV C C/N 012 was not formally published, but approved changes have been incorporated in REV D.	RFC N01-0644 CCB 0013-18 dtd 22 Mar 2018
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NOTE: Added requirements are identified with an NGA Effectivity number (NEn) in the right margin.

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1. (U) INTRODUCTION

1.1 (U) PURPOSE

(U) The National System for Geospatial Intelligence (NSG)/Allied System for Geospatial-Intelligence (ASG) Enterprise Test and Evaluation Master Plan (TEMP) provides a master plan for independent test and evaluation (T&E) of all changes to the NSG/ASG Enterprise prior to operational deployment. The NSG/ASG Enterprise TEMP describes how the National Geospatial-Intelligence Agency's (NGA's) independent test program provisions, manages, and operates test resources and the organizational roles and responsibilities. The NSG/ASG Enterprise TEMP provides information about advanced and non-traditional technologies, development approaches and systems, and traditional and hybrid development approaches.

1.2 (U) SCOPE

(U) The NSG/ASG Enterprise TEMP describes the independent T&E activities required to support the integration of sensor data, hardware baselines, and software baselines across NGA and the user communities. New requirements and identification and correction of errors and deficiencies drive changes to the applications, services, sensor data, and systems that compose the NSG/ASG Enterprise. These changes require frequent testing. The T&E process is tailorable to support a variety of development approaches, including waterfall, spiral, agile, Development and Operations (DevOps), mobile applications, Application Program Interfaces, and hybrids.

(U) The NSG/ASG Enterprise TEMP references software, sensor data, hardware, systems, applications, services, widgets, desktops, corporate systems, or networks that require testing. For clarity and ease of reference, "entity" will be used to represent those terms unless noted otherwise. The term "test plan" refers to an NTO or NIQU test plan unless stated otherwise. Additionally, the terms "test," "test event," and "assessment" are used interchangeably; they are all meant to describe the activities occurring during the test execution and are not intended to convey separate activities.

(U) The NSG/ASG Enterprise TEMP describes the comprehensive T&E strategy for independent testing of the NSG/ASG Enterprise, both present and future.

(U) NGA does not require programs or projects to produce a separate government TEMP or an appendix for this NSG/ASG Enterprise TEMP for their activities. During the development phases of a program or project, development contractors should follow the test processes of their companies; many contractor organizations have an internal TEMP.

(U) Although the test process is adaptable, basic principles and criteria are essential to producing valid test results. As NGA has pursued a more agile development approach to reduce system complexity and deployment time, test processes and strategies have been improved to support accelerated release cycles.

1.3 (U) MISSION DESCRIPTION

(U) Enterprise-level testing is conducted by the Integration Test & Image Quality Office's (TAC's) Assessment Division (TACA) and Image Quality Division (TACQ) to provide an independent quality assessment of products provided to the user community to meet mission needs. TACA is also referred to as the NGA Test Organization (NTO); TACQ is also referred to as the NGA Image Quality and Utility (NIQU).

(U) Independent testing is conducted using the NTO Test Environment (NTO TE), NIQU Evaluation Facility (NEF), NIQU Engineering Cell (NEC), and assets at associated facilities. The NTO TE comprises dedicated test resources in two locations: NGA Campus East (NCE) and NGA Campus West (NCW). The NEF and NEC are specialized assessment environments located at NCE. "Associated facilities" refers to other sites where capabilities are located that are routinely used to support or participate in testing through a network connection. In rare circumstances, it may be necessary to conduct testing at additional sites. Such circumstances require prior approval by TAC management.

(U) The focus of an independent test program is to support preservation of the integration and interoperability of the NSG/ASG Enterprise mission production information technology enterprise. To do this effectively, test resources are allocated to those applications, services, systems, sensor data, or infrastructure items that pose the greatest risks to integration and interoperability. Identification of entities requiring test is a shared responsibility of Program, Project, or Product Managers (PMs) and TAC and is based on the risk that the entities pose to other NSG/ASG Enterprise components or user workflows.

The Director of the Integration Test & Image Quality Office (D/TAC) of the Transformation Services Group (TA) is the approval authority for the NSG/ASG Enterprise test process.

1.4 (U) CONFIGURATION MANAGEMENT CONTROL

(U) This document is controlled by the NGA Configuration Control Board (NGA CCB). The NSG/ASG Enterprise TEMP will be modified when the D/TAC determines that the existing NSG/ASG Enterprise TEMP no longer provides adequate guidance to cover the NGA T&E process. All changes to this document shall be coordinated via the Request for Change (RFC) process and approved by the NGA CCB prior to incorporation.

1.5 (U) CRITICAL SYSTEM CHARACTERISTICS

(U) Universally applied critical system characteristics are called key performance parameters (KPPs) and are described in NGA's Operational Requirements Document (NORD). Required capabilities are described in a Capability Development Document (CDD). Critical technical parameters (CTPs) are derived from the KPPs or the CDD associated with a particular program or project, internal NGA Enterprise system, or contractual agreement. Due to the general nature of KPPs, under most

circumstances, TAC will use specific capabilities and requirements documentation that trace back to the KPPs in order to understand new required capabilities and to develop test objectives to verify and validate them. Examples of requirements documentation include the Product Quality Operational Readiness Criteria (PQ-ORC), applicable segment/specification requirements documents (SRDs), segment/system requirements verification traceability matrices (RVTMs), user concept of operations (CONOPS), features, and feature acceptance criteria.

2 (U) TEST STRATEGY

2.1 (U) INTRODUCTION

(U) T&E is a process that spans the full lifecycle of a system. The Defense Acquisition System defines several categories of testing, three of which apply to NGA acquisition programs: Developmental Test and Evaluation (DT&E), Operational Test and Evaluation (OT&E), and DevOps.

(U) DT&E is conducted early in a system's lifecycle and is focused on ensuring a system's design meets its requirements. DT&E plans for software development document a specific progression of testing: unit testing, component testing, and system testing. For NGA acquisitions, DT&E is conducted by the developer; however, tests conducted near the end of DT&E are executed under close government program management office (PMO) supervision. Formal requirements verification is accomplished by the government's PMO during one of the closely supervised tests. One of TAC's goals during DT&E is to integrate with the developer by collaboration, to identify issues early in the development pipeline, and provide testability assessments for feature/capability acceptance criteria and requirements.

(U) OT&E is conducted late in a system's lifecycle and is focused on independently verifying that the delivered system or sensor data performs effectively and is suitable for use in its intended operational environment. OT&E is conducted exclusively by the government with the assistance of independent contractors and the user community; Department of Defense (DoD) strictly limits development contractor involvement to ensure the integrity of test results. One of TAC's goals during OT&E is to ensure the correct implementation of an entity's design and its suitability for use in its intended operational environment.

(U) NGA is developing a DevOps strategy. The DevOps process is a foundational element to digital transformation, and it allows for collaboration among developers and assessment teams. It also allows organizations to identify issues earlier in the development pipeline. This, along with continual integration and testing, reduces time spent troubleshooting and improves average time of repair. Higher-quality products are therefore delivered to the end users more quickly and more frequently. One of TAC's goals during the development to operations stage is to ensure that the developer is using testing best practices that help improve the quality of the products deployed to the end user. TAC will be a collaborative partner in the effort among the developers, product owners (POs), and the end user, to ensure quality products are delivered in the most efficient and effective way possible.

(U) TAC supports the mission of the Geospatial Intelligence (GEOINT) Functional Manager Standards Assessment (GFMSA) Program by ensuring that GEOINT capabilities, segments, and software developed for deployment into the NSG/ASG adhere to mandated interoperability and quality standards. TAC applies early engineering support and rigorous testing to all GEOINT entities to

facilitate their seamless integration and to minimize any negative impacts to the NSG/ASG.

(U) TAC testing is expected to be both realistic and rigorous, while striking a balance between cost (time and resources) and risk. The test program is also adaptable to new technologies (e.g., Internet of Things (IoT) solution space, Microservices) and innovative development approaches (e.g., agile development, DevOps).

2.2 (U) GENERAL METHODOLOGY

(U) TAC conducts most of its testing in a controlled test environment that closely simulates the operational environment in which the entities will function (e.g., data center, cloud). The fidelity of this controlled test environment is enhanced by the use of operational data (e.g., imagery datasets and other products) during testing. Simulated test data (also known as engineering data) is used only when operational data is unavailable.

(U) Most TAC testing employs a thread-based day-in-the-life (DITL) test approach, in which testers verify a new system for product quality (PQ), security, interoperability, and functional integration. Testers also verify the output, by executing typical user procedures to cause the system to perform the specific business processes or workflows it was designed to support. The procedures are selected to ensure all inter-system interfaces are exercised in order to confirm that the new development has been successfully integrated into the operational architecture and that it does not interfere with the functioning or stability of any other systems with which it operates. Because of the flexibility thread-based DITL testing offers, TAC uses thread-based DITL testing for the majority of its test activities, regardless of the specific type of development to be tested (e. g., traditional segment, web-based service, desktop application, or network).

(U) Testing in a controlled environment that closely replicates the operational environment is the most effective way to identify issues that could adversely affect the user and mission. Implementing a risk management strategy that focuses on the areas that pose the greatest risks increases the efficiency of resources. This approach is referred to as Risk-Based Testing (RBT).

(U) NGA strives to develop and field new and improved capabilities on a timeline that keeps pace with the end user's mission requirements. NGA is increasingly implementing agile software development approaches to decrease the time between identifying and satisfying a need. Current NGA agile development contracts are using a variety of design approaches, development approaches, and delivery rhythms. TAC continues to adapt its processes and align its resources to support agile program requirements while accomplishing its primary function of preserving the integrity of the current systems in the NSG/ASG operational baseline. TAC has created a repeatable agile testing approach to

complement the cadence of an agile program. TAC's strategy is to manage its resources to accommodate various development approaches and to provide the highest quality test support possible.

(U) External entities may be accepted for integration into the NSG/ASG environment, if TAC determines that the test results and test processes from the external agency were acceptable. TAC will review all test artifacts pertaining to the external entity and will determine if additional functional integration, interoperability, security, or image quality testing is needed. TAC will also review all test artifacts pertaining to an external organization's security testing and will accept the segment for integration, if the test processes are equivalent to TAC's security testing.

2.3 (U) TEST SUPPORT FUNCTIONS ROLES AND RESPONSIBILITIES

2.3.1 (U) Business Area Lead

(U) A business area lead (BAL) is designated for each NTO and NIQU business area. The roles and responsibilities for both NIQU and NTO BALs (unless specifically stated otherwise) are addressed throughout this NSG/ASG Enterprise TEMP; key roles and responsibilities include the following:

- (U) Review proposed test strategy and test plan for technical accuracy prior to approval
- (U) Review test objectives and plans for technical clarity, completeness, and achievability
- (U) Provide technical guidance on testing methodologies
- (U) Approves test requests and projects in Verification and Validation On-Line Tools (VOLT) for all activities in their areas of responsibility
- (U) NTO: Conduct a post-test review to ensure that the test objectives were met and that the recommendation to the Program/Project Manager (PM) is clearly stated
- (U) NIQU: Conduct a post-test review to ensure that the test objectives were met and that deliverables clearly document the results of the effort
- (U) NTO: Share information at team meetings
- (U) NIQU: Represent NIQU at NTO planning meetings
- (U) NIQU: Coordinate with the Ops Lead(s) to ensure sufficient resources and schedule are available to complete the topic

2.3.2 (U) NTO Test Director

(U) A test director (TD) is designated for each NTO full-service test activity. The TD's roles and responsibilities are addressed throughout this NSG/ASG Enterprise TEMP; key roles and responsibilities include the following:

- (U) Establish or lead meetings as appropriate (e.g., Test Integrated Product Team (IPT) meetings, daily start-up meetings, daily wrap-up meetings)
- (U) Coordinate with PM(s) to understand the entity under test and devise a collaborative test strategy
- (U) Provide advice and best practices to help ensure delivery of quality products
- (U) Attend test readiness reviews (TRRs) to provide test readiness status
- (U) Review and comment on RFCs and Engineering Change Proposals (ECPs) for test impacts and adequacy of test planning and test resources
- (U) Engage with the PM(s) on all design issues in order to better understand how to independently test capabilities being delivered to operations
- (U) Engage with the user community and other stakeholders on all issues relating to system CONOPS and user/mission workflow in order to better understand how to independently test capabilities being delivered to users in operations
- (U) Coordinate with NIQU topic leads to ensure integration of NTO and NIQU testing, where appropriate
- (U) Develop test objectives for test activities
- (U) Validate test objectives (during test execution) and revise objectives as appropriate
- (U) Write test plans, daily status reports, and final status reports
- (U) Coordinate with necessary stakeholders to ensure the availability of the necessary test environment and other test support services
- (U) Execute tests and ensure compliance with approved test processes

2.3.3 (U) NIQU Topic Lead

(U) The NIQU topic lead is responsible for the successful execution of a test. The topic lead's roles and responsibilities are addressed throughout this NSG/ASG Enterprise TEMP; key roles and responsibilities include the following:

- (U) Develop and maintain a detailed task list, execution plan, and schedule
- (U) Coordinate with internal and external partners and customers
- (U) Execute the topic according to the approved plan and NIQU processes
- (U) Provide status updates as required
- (U) Coordinate the scheduling of all topic-related briefings and meetings

- (U) Schedule and conduct all necessary meetings with the topic team
- (U) Schedule and conduct a topic kickoff meeting to communicate to the team the schedule, assignments, and deliverable requirements associated with each task
- (U) Schedule and conduct readiness reviews in accordance with the NIQU readiness process
- (U) Promptly report and document any risks and issues associated with fulfilling scheduled deliverables to the NIQU BAL
- (U) Ensure topic tracking data is captured in the NIQU portal and monitor any deviations from approved plans
- (U) Coordinate with NTO TDs to ensure integration of NTO and NIQU testing, where appropriate

2.3.4 (U) Program Manager/Project Manager and/or Product Owner

(U) The PM or PO responsibilities include the following:

- (U) Budget and contract the T&E need of the program throughout program lifecycle
- (U) Align appropriate type of T&E activities according to the program's lifecycle
- (U) Manage developmental testing for the product
- (U) Support TAC T&E activities, as identified in consultation with the TAC testing's point of contact (POC)
- (U) Work with the TD and, if applicable, the Test IPT to manage the test strategy
- (U) Submit Requests for Services (RFSSs) to obtain test support
- (U) Include in the entity budget the funding to cover the cost of providing TAC with a suitable suite of hardware (physical or virtual) and fully licensed version of the software to support testing of the delivered entity for the life of the entity, unless a waiver is obtained from the D/TAC
- (U) Include in the product budget the funding necessary to provide full-time, on-site product engineers to work in the NTO TE to provide all system administration and testing support for all activities pertaining to the entity's hardware, software, or interfaces for the life of the entity, unless a waiver is obtained from the D/TAC
- (U) Provide required documents (CONOPS and SRDs) and other relevant documentation as required (e.g., requirements traceability; operational scenarios; factory acceptance test (FAT) procedures; FAT results; test data; training, testing, and security documentation) to support TAC test

planning and execution and the Certification to Field (CtF) process

- (U) Coordinate with the TD to ensure development contractors provide adequate T&E support
- (U) When applicable, ensure appropriate members of the entity staff regularly and actively participate in the Test IPT
- (U) Develop security test plans and procedures in accordance with the governing Director National Intelligence (DNI) and Department of Defense Intelligence Information System (DoDIIS) requirements
- (U) Provide approved security test plans and procedures specific to each NTO test plan (each security test plan will be submitted to Cyber Security (TSR) for formal approval)
- (U) Certify that the product has completed formal requirements verification, has successfully completed FAT (or equivalent testing) or a Pre-Ship Review and is ready for integration and operational testing
- (U) Arrange for NTO test engineers to be trained on new entity capabilities prior to NTO testing
- (U) For those systems requiring certification, provide input for the draft and final certification package to the NTO Intelligence Information Support Team (NIIST)
- (U) Obtain National Image Transmission Format (NITF) or other necessary certifications prior to TAC testing
- (U) Evaluate the entity for potential PQ risks and, if any, indicating the need for PQ testing in VOLT
- (U) If appropriate, coordinate with NIQU for PQ testing during entity development to obtain early feedback on program compliance with applicable PQ standards and/or NORD KPPs 1, 3, and 4
- (U) Provide NTO with two copies (Gold Copy) of the software baseline (if delivered on media) and a list of all commercial off-the-shelf (COTS) and government off-the-shelf (GOTS) products being delivered to the NTO TE
- (U) Provide updated product baseline to the TAC Configuration Management (CM) Library, through TAC CM, upon receiving successful testing results

2.3.5 (U) Joint Interoperability Test Command

(U) The Joint Interoperability Test Command (JITC) responsibilities include the following:

- (U) Provide personnel, co-located within TAC, to assess early insight into entity developments that will require JITC evaluation

- (U) Monitor and validate the interoperability of entities being tested by TAC during Independent Verification and Validation (IV&V) testing

2.3.6 (U) Infrastructure Service Provider

(U) The Infrastructure Service Provider (ISP) responsibilities include the following:

- (U) Providing and maintaining the collective NTO TE, including the current desktop test environment, the thin client test environment, the NCE virtual test environment and the cloud test environment (NTO/NIQU determines the configuration of the NTO and NIQU test infrastructure, including networks and the mechanism for coordinating changes to the test environments and obtaining NTO or NIQU approval is the RFS, submitted via VOLT)
- (U) Provisioning and maintaining servers and desktop workstations to support any architecture in use by NGA (the concept for ISP maintenance of the test environments is the same as for operations, to include recapping of hardware as part of the recap schedule for operations; ISP maintenance also includes provision of the same upgrades of operating systems, server and database software, and other items as provided by the ISP to both mission and corporate operations; ISP provision of ISP software and software upgrades includes licenses, except for test tools)
- (U) Test instantiations of new ISP hardware and software are provided for the test environment prior to deployment to the operational environment so that they can be tested and retained for future testing.
- (U) Obtaining test support for ISP hardware and software, including the NGA desktop infrastructure, the ISP organization sponsoring or making the change submits an RFS via VOLT to request NTO test support
- (U) For desktop infrastructure changes, the Core Service Division of NGA's Service Implementation Group provides a test package that includes fundamental data element information, installation and configuration procedures, the Unit Test Report, and other applicable information, to assist in developing appropriate test objectives for the independent test.

(U) NTO's ability to continue providing effective test support to NGA and the NSG/ASG depends heavily on the ISP ensuring the NTO test infrastructure remains current with operations.

2.3.7 (U) Cloud Service Provider

(U) The NGA Cloud Integration & Adoption (NCI&A) PMO is empowered to work across directorates to establish unified, transparent, and accountable processes and procedures and to provide the full complement of services needed by NGA and its customers for Cloud operations. The NCI&A PMO's responsibilities

include the following:

- (U) Provide oversight and guidance detailing architecture and design principles, as well as standards needed for technical conformance enabling Intelligence Community (IC) Enterprise vision and mission realization
- (U) Coordinate, advocate, facilitate, and enable Federal and DoD Cloud initiatives and integration across NGA and in partnership with the IC
- (U) Execute NGA's required activities to transition to a hybrid Cloud computing architecture operational within IC ITE, Government Cloud Services (GovCloud), and Commercial Cloud Services (C2S), or in compliance with the DoD Cloud Directives
- (U) Establish a procedure to address NGA-required activities for transitioning to a Cloud computing architecture operational within the Intelligence Community Integrated Test Environment (IC ITE)

2.3.8 (U) Consolidated Systems Engineering Review Board

(U) The Consolidated Systems Engineering Review Board (CS ERB) responsibilities include the following:

- (U) Reviewing, prioritizing, assigning, and adjudicating all discrepancy reports (DRs) identified during post-FAT integration, test, and transition activities
- (U) Reviewing, prioritizing, assigning, and adjudicating all inter-segment DRs for those elements that have not been transferred to operations
 - Inter-segment DRs identified during the development phase
 - All DRs identified during the conduct of system-level integration or site acceptance test and transition activities
 - Inter-segment DRs identified during operations for those elements that have not been transferred to operations.

2.3.9 (U) Applications Services Governance Board

(U) The Applications Services Governance Board (ASGB) is a forum that acts as the NGA's central governing body for the NGA GEOINT App Store and other similar Programs of Record (PORs). A Terms of Reference (TOR) document for the ASGB, approved and coordinated with all NGA directorates and key staff offices, articulates the centralization of governance as it relates to mission value, prioritizations, and policy recommendations for application deployment to the NGA GEOINT App Store. However, the TOR does not discriminate or preclude other PORs from using this board for vetting and approving purposes. ASGB responsibilities include the following:

- (U) Adjudicate or demote applications, web services, data, and any other

NGA development

- (U) Promote into operations any application, web services, or data

(U) ASGB membership includes voting representatives from directorates across NGA, including:

- Chief Information Office (CIO) and Information Technology Services (CIO-T)
- Office of the General Counsel (OGC)
- Analysis (A) Directorate
- Research (R) Directorate
- Source (S) Directorate
- TAC
- TS (Security and T&E)

3 (U) INTEGRATED TEST PROGRAM

3.1 (U) INTRODUCTION

(U) This section describes NGA's integrated test program in terms of the three major categories of testing and the service offerings in each category. The NSG/ASG Enterprise TEMP presents these categories as composing a well-defined sequence of specific services that an entity is subjected to as it progresses from concept to deployment. However, because the Defense Acquisition System and the test and evaluation strategy emphasize the integration of test activities, the test sequencing is adapted to fit the needs of systems being developed in new development processes (i.e., DevOps), even for systems being developed under traditional acquisition approaches. As NGA adopts new acquisition strategies and development approaches, TAC expects to exploit opportunities for greater integration of the TAC testing services. Therefore, the reader should bear in mind that although the sequence of specific service events presented in this section is useful for instructional purposes, it is not to be interpreted as a rigid prescription for the design and implementation of any specific entity's test and evaluation plan.

(U) Appendix A contains a tabular summary of the various types of NGA's integrated test program types discussed in this section and relates them to their Defense Acquisition System test categories.

3.2 (U) SUPPORT TO DEVELOPER TESTING

(U) As mentioned previously, the Defense Acquisition System defines two broad categories of testing that apply to NGA: DT&E, and OT&E. Figure 3.1 illustrates the major test events that compose these two categories as implemented by NGA.

3.2.1 (U) Developmental Test and Evaluation, Phase 1

(U) Figure 1 depicts DT&E services as consisting of two phases that are primarily conducted by the development contractor under varying degrees of government PMO or PO supervision. This phase of testing begins with unit testing of individual modules of code and culminates in the conduct of a FAT for the final delivery. It is during FAT or an equivalent activity (such as a product demonstration of new capability) that developers accomplish formal requirements verification under government PMO or PO supervision. This includes verifying compliance with any required standards (e.g., NITF). Although most of the first phase of DT&E testing is internally focused, developers may choose to coordinate and conduct early interface tests (EITs) with other developers to provide an opportunity to assess a portion of their products' capabilities as they interoperate with one or more external entities.

(U) The planning and execution of the first phase of DT&E testing is the responsibility of NGA PMOs or POs and their development contractors and is not governed by the provisions of this NSG/ASG Enterprise TEMP. However, NTO

TDs usually attend EITs, FATs, and product demonstrations as observers to gain familiarity with new systems and to better understand the entity-specific requirements for subsequent independent testing. NIQU also uses EITs, FATs, and product demonstrations as opportunities to provide development contractors with early feedback on the format compliance of their products, insight into systems' compression implementation, and in some cases, assessment of the accuracy of the support data with which the products are being populated. NTO's and NIQU's early engagement with PMO/PO reduces risk during later formal entity testing.

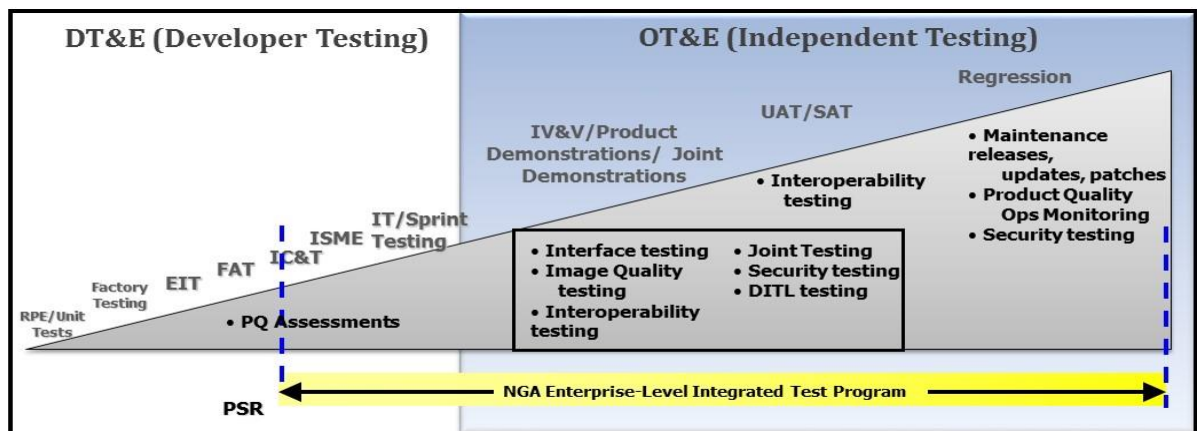
(U) In the traditional acquisition process for NSG/ASG mission capabilities, the event normally marking the transition from the first phase of DT&E to the second phase of DT&E is the pre-ship review (PSR). Following a successful PSR, all DRs identified as open at the PSR or written subsequent to the PSR are entered into the NGA DR system (Remedy) for tracking and adjudication by the appropriate engineering review board or configuration management board.

3.2.2 (U) Developmental Test and Evaluation, Phase 2

(U) Because developer test environments do not include direct interfaces with all of the NGA entities they integrate with, developers will generally not be able to fully verify successful integration of their products with the operational NSG/ASG enterprise architecture. Therefore, the second phase of DT&E is typically conducted at the NTO TE or another NTO-approved facility. It begins with the delivery of the new or enhanced entity to the NTO TE (or other approved site) where developers perform installation, checkout, and test (IC&T) of their entities to ensure that they have been properly configured and can communicate with all applicable systems in the test environment. The NTO and NIQU test engineers will observe and provide the PMO/PO with a facilitator to assist in any environment related issues. Developers or PMO testers are then afforded the opportunity to evaluate the proper integration and functioning of their entities by leading and conducting one or more integration tests (ITs). The second phase of DT&E concludes with the completion of the last IT, at which time the PMO/PO turn the entities over to the NTO for independent, enterprise-level OT&E.

3.3 (U) GOVERNMENT TESTING

(U) The T&E that occurs after the second phase of DT&E is referred to in this NSG/ASG Enterprise TEMP as "government test," which is equivalent to the Defense Acquisition System category of OT&E. The earliest and most methodical government test events are those conducted by TAC (NTO in collaboration with NIQU, OCIO, and JITC). If applicable, after this early testing has confirmed that the product is stable and functional and that it poses no risk to the operational architecture, user organizations conduct the remaining OT&E events, usually with the assistance of the sponsoring PMO at a specific site location. The government testing that is conducted at the NTO TE can be combined with user testing when the users are included in the various test and evaluation service offerings that make up the specific T&E strategy for a PMO.



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Figure 1. (U) NGA Enterprise-Level Integrated Test Program

3.4 (U) QUALITY ASSURANCE SERVICES

(U) This section describes, at a high level, the quality assurance (QA) services provided by the divisions within TAC. Those services fall into six categories:

- Engineering Analysis
- Test Environment Hosting
- Test Execution
- Readiness
- Security Assessments and Scans
- Business Process Management

3.4.1 (U) Engineering Analysis

(U) The Engineering Analysis service is the application of scientific and analytic principles and processes to identify the properties and condition of the entity or GEOINT data under review. This service contains studies, requirements analyses, operational issue investigations, quick look assessments, site acceptance tests (SATs), standards and scales development, consultations, and performance profiling.

3.4.1.1 (U) Studies

(U) The Studies subservice provides an analysis that is performed through research or observation of the entity or data to understand the behavior of the entity. Through this subservice and the analysis performed in this subservice, TAC may identify the need to conduct a small-scale preliminary assessment to evaluate feasibility and performance. This analysis is to validate a design or desired outcome of a technological update with respect to sensor data quality

(e.g., the cloud environment) and modifications to functionality, interoperability, sensor data quality, and utility requirements for systems and data.

3.4.1.2 (U) Requirements Analysis

(U) The Requirements Analysis subservice provides an analysis of the requirements to ensure that the GEOINT capabilities being delivered provide sensor data quality, interoperability, performance, utility, and data integrity. Through this subservice, TAC provides the PMO with an independent review of relevant requirement documents.

3.4.1.3 (U) Operational Issue Investigations

(U) The Operational Issue Investigations subservice provides support for resolution of operational sensor data quality discrepancies related to processing, exploitation, or dissemination.

3.4.1.4 (U) Quick Look Assessments

(U) The Quick Look Assessments subservice provides abbreviated assessments (e.g., fewer assessors, datasets, activities, or time) conducted to provide insight into the behavior of an entity or data type under specific circumstances.

3.4.1.5 (U) Site Acceptance Test

(U) The Site Acceptance Test (SAT) subservice provides users the opportunity to gain hands-on experience with a new capability, allowing them to qualitatively assess how well the capability meets operational needs and expectations, and to assess interoperability, security attributes, and performance before transition to operations. SATs occur on servers outside of the NTO TE (i.e., in operations), which TAC does not control. TAC personnel may observe, participate in, or facilitate (but not lead) the testing. TAC resources (e.g., workstations, subject matter experts (SMEs)) may be used to support testing.

(U) An SAT follows successful completion of all other government testing. It may include aspects of entity testing such as functionality, interoperability, PQ, security, and performance testing.

(U) An SAT can be as formal as desired by the user organization. For a well-understood system with few changes, the SAT may simply consist of ad hoc exercising of the product by one or more users. For new entities or major upgrades, the user organization may elect to have a fully scripted test to ensure all major parameters are exercised and to develop user confidence in the new capability. Users may conduct the SAT with development contractors available to provide SME guidance.

3.4.1.6 (U) Standards and Scales Development

(U) The Standards and Scales Development subservice provides the documentation of processes and procedures that are necessary for the NSG/ASG to be compliant with NORD KPPs 1, 3, and 4. The development of quantifiable measures help to define the levels of interpretability, utility, and

interoperability provided by GEOINT sensor data.

3.4.1.7 (U) Consultations

(U) The Consultations subservice provides scientific, engineering, statistical, or technical interactions on hardware, software, processes, methodologies, and innovations.

3.4.1.8 (U) Performance Profiling

(U) The Performance Profiling subservice provides analysis of application performance to identify potential bottlenecks or support application optimization. This subservice may include ongoing performance audits to evaluate application performance against established benchmarks.

3.4.2 (U) Test Environment Hosting

(U) The Test Environment Hosting service is the provision of test environment resources to test entity capability across interfacing test systems; the PMO or developer is responsible for conducting the test. This service contains Inter-Segment Message Exchanges (ISMEs), EITs, Enterprise Tests (ETs) and System Demonstrations (SDs), IC&Ts, ITs, Assisted-Service IV&Vs, and Assisted-Service Regression Tests.

(U//FOUO) Government testing of NGA mission capabilities includes joint operational testing or demonstrations with external organizations that interface with the NGA enterprise architecture. Joint testing has essentially the same objectives as internal NGA testing: to ensure that modifications accurately respond to formal requirements and that they will not adversely affect the joint operational environment. The National Reconnaissance Office (NRO) is the NGA's primary mission partner; the types of testing required for various types of changes to the joint NRO/NGA enterprise are documented and coordinated via the joint RFC process and several joint planning forums. When changes to the NRO side of the enterprise are planned, the NRO test program drives the types of test activities and the schedule, with TAC participating. Conversely, when changes to the NGA side of the enterprise are planned, TAC drives the types of test activities and the schedule, with the NRO participating. When major changes to the NRO components of the enterprise drive the need for joint test, the standard phases of the NRO test program that may require TAC participation are the EIT and the ET. However, because the NRO manages the testing of its legacy systems under an earlier naming convention, the EIT may be referred to as an ISME and the ET may be referred to as an SD, a Stability and Operability Test (S&OT) or a Transition/Operations Confidence Test (T/OCT).

3.4.2.1 (U) Inter-Segment Message Exchange

(U//FOUO) The ISME subservice provides a hosted test environment to enable the NRO to assess interfaces via message exchange that could not be tested in the factory test environment.

3.4.2.2 (U) Early Interface Test

(U) The EIT subservice provides a hosted test environment to enable the developer to assess NSG/ASG interfaces that could not be tested in the factory test environment.

3.4.2.3 (U) Enterprise Test and System Demonstration

(U//FOUO) The ET and SD subservice provides a hosted test environment to evaluate the compatibility of NGA and other mission partners' system interfaces via message exchange and DITL testing.

(U) In response to a spacecraft or sensor anomaly, NGA participation may involve anomaly work-off, Return to Mission (RTM) certification and joint Enterprise Initialization.

(U) When major NGA changes are the primary reason for the test series, the test schedule organization is based on the NGA model described in Sections 3.2.1 through 3.3. This is followed by a joint ET to validate the continued operability of the joint enterprise with the addition of the new entity.

3.4.2.4 (U) Installation, Checkout, and Test

(U) The IC&T subservice provides a hosted test environment to enable developer assessments of software installations and system readiness for test. Upon delivery to the NTO TE or another agreed-upon location, the developer performs IC&T to configure product hardware, install software, and conduct initial checkout and testing to ensure that the entity is prepared for more detailed and comprehensive testing. The NIIST may use this opportunity to validate the documented installation procedures provided by the developer; this formal validation activity may be conducted later by re-installing the entity as coordinated between the NIIST and the developer. The NIIST may either execute the install procedures or observe the execution of the procedures by the developer. The objective is to ensure that the documented procedures are fully executable and sufficient to correctly install the entity. Success criteria for an IC&T consist of successfully completing installation of hardware and software, communications checks, database initializations, and message exchange tests to verify operation of the entity. For many systems, installation is performed via a factory push. In these cases, there are no installation procedures to validate. Instead, the NIIST will record when the push occurred for configuration management.

3.4.2.5 (U) Integration Test

(U) The Integration Test (IT) subservice provides a hosted customer-led test event to assess the entity's ability to be integrated into the NSG/ASG Enterprise baseline without adversely affecting other systems. The ITs assure the customer that component systems have been successfully integrated, that the interfaces function correctly, and that the entity is ready for formal testing in the enterprise infrastructure. After IC&T for a major new delivery, the developer and PMO/PO

conduct ITs in preparation for formal testing. In most instances, a minimum of two ITs is recommended as an important risk mitigation factor before the entity moves to IV&V. The NTO normally provides the test environment and a TD to facilitate and provide guidance on the NTO procedures and policies to the PMO/PO and developer in performing these tests. However, preparation of the test plan and procedures and the actual conduct of the test are the responsibility of the sponsoring government PMO/PO. In most cases, this will be the first opportunity for the entity to operate with all of its actual interfaces, which often leads to the detection of issues that may not have been possible to detect at the developer's facility. For this reason, developers have full control of their software during ITs so that they may correct and retest their deliveries without being constrained by external configuration management.

3.4.2.6 (U) Assisted Service Independent Verification and Validation

(U) The Assisted Service IV&V subservice provides a hosted test environment that provides a controlled operationally representative environment to enable customers to conduct testing of major upgrades (e.g., new functionality) to their entities. This subservice is used for systems that have been assessed as mature in their development and testing processes.

3.4.2.7 (U) Assisted Service Regression Test

(U) The Assisted Service Regression Test subservice provides a hosted test environment that provides a controlled operationally representative environment to enable customers to conduct testing of minor upgrades (e.g., DR fixes) to their entities.

3.4.3 (U) Test Execution

(U) The Test Execution service provides verification and validation that an entity meets its intended purpose with regard to functionality, quality, interoperability, and utility. This service contains IV&Vs, sensor initializations (joint), agile tests, web services tests, regression tests, emergency tests, Information Assurance Vulnerability Alert (IAVA) Tests, and operations assessments.

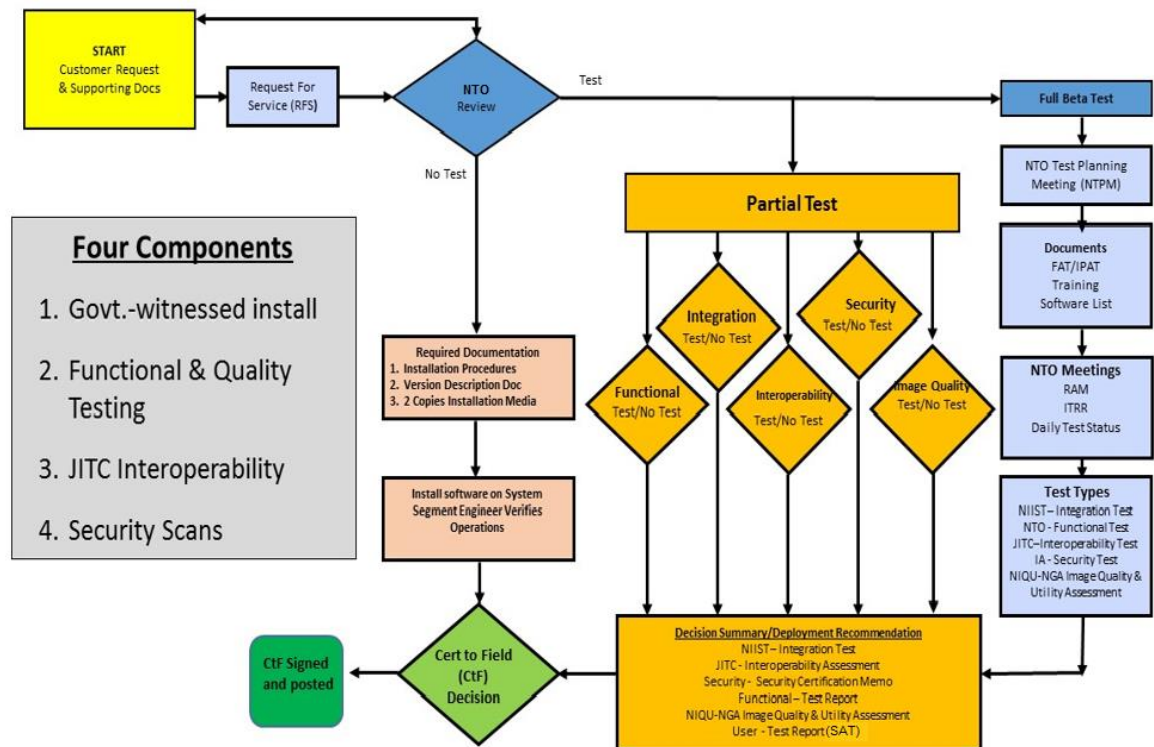
3.4.3.1 (U) Independent Verification and Validation

(U) The IV&V subservice provides a verification that an entity meets its intended purpose with regard to functionality, quality, interoperability, and utility. The objectives of IV&V are to verify that the tested entity can be successfully integrated into the operational baseline (i.e., is stable, does not interfere with the functioning of other systems, and successfully performs all inter-system message exchanges) and to validate its inter-system functionality. TAC uses DITL testing focusing on data flows and functional/capability threads or "business processes" that cross inter-segment interfaces to accomplish these objectives. IV&V does not duplicate the verification of segment requirements that should have been verified by PMOs during FAT or an equivalent test event. However, if a program office is unable to verify some requirements during FAT due to test environment

limitations at the development facility, these requirements can be identified at PSR (or an equivalent event) as liens against the system to be verified during independent testing in the NTO TE, NIQU test facilities during IV&V, or at an operational site during SAT. Following a successful IV&V, the tested baseline remains frozen until operational deployment. Baseline changes made after operational deployment follow normal configuration management control processes.

(U) The NGA CtF procedure is designed to ensure the quality and integrity of the NSG/ASG entities delivered to the NSG/ASG Enterprise. This procedure is based on the DoDIIS Certification Process and entails a disciplined, robust, and repeatable IV&V and evaluation process to ensure that critical quality parameters are met for the following:

- Integration: Conducted by NIIST; documents procedures based on DoDIIS-defined integration requirements and commercial best practices.
- Security: Conducted by the Software Assurance Team; validates compliance with System Security Authorization Agreement (SSAA) and System Security Plan (SSP). Tests and certifies the security aspects of the system in support of security accreditation for fielded NGA products. Test activities are structured to allow the certification tests to occur immediately after DoDIIS integration tests to meet certification timelines.
- Interoperability: Conducted by JITC representatives collocated with the NTO, who observe the interoperability of the test configuration with other DoDIIS, Joint, and Service systems and identify the functional interoperability requirements to be executed as part of NTO testing. They work with the test director and NIIST lead to identify any specific interoperability test cases to be included in the test events. JITC representatives develop and execute interface-specific test plans and observe users conducting interoperability testing. This testing is conducted during IV&V testing and performed for systems being deployed both internally within NGA and externally to Major Commands and Services.
- Functionality/Capability: Conducted by the NTO; provides integration and functionality/capability testing of interfaces based on DITL scenarios. Testing is conducted by independent testers (operational users, organizational Conflict of Interest-free contractors, government personnel); functional/capability test activities are coordinated by the government PM.
- Image Quality: Conducted by NIQU; image quality testing is performed as part of IV&V testing. NIQU samples and technically analyzes sensor data to ensure that metadata integrity and pixel quality has not been adversely affected by the new entity being tested. A detailed explanation of the processes followed for image quality testing is provided in Section 5.8.

(U) Certification-to-Field Decision Process

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Figure 2. (U) Certification-to-Field Decision Process

(U) Appropriate certification authorities (Integration, Security, Interoperability, Functionality/Capability, NIQU) are engaged early in the test planning effort to identify appropriate windows in the test schedule where certification activities should be accomplished. The NIIST collects the recommendations from the test events and provides the NGA CtF document.

(U) The NTO conducts six enterprise desktop build IV&V test activities throughout the calendar year. Enterprise desktop changes are initiated by a Remedy change request (CR) and managed by NGA's Core Service Division of the Information Technology Services Directorate's Service Implementation Group. The Engineering Work Group (EWG) provides information to NGA stakeholders regarding baseline changes in work by the Core Service Division and the availability of the software package to be provided by the Software Engineering Integration Team (SEIT). The information provided by the EWG and the RFS assists the NTO in determining the test strategy for the upcoming enterprise desktop test.

3.4.3.2 (U) Sensor Initialization (Joint)

(U) The Sensor Initialization (Joint) subservice provides a review of data from

new and existing sensor families for sensor performance. This assessment focuses on geolocation and mensuration accuracy, support data and visual rendition accuracy, or interoperability and functional performance throughout the Enterprise using an end-to-end data flow. All activities are used to determine readiness to transition to operations.

(U//FOUO) System PQ sensor initialization covers the full extent of testing performed to verify that a new enterprise imaging sensor is ready to be transitioned to operations. Initialization testing includes the PQ test techniques described in Section 3.4.5. These techniques are performed in a coordinated manner and scaled to fit the testing schedule of the task. Initialization testing is done in close cooperation with the mission partner and is reported on a daily basis to appropriate levels within NGA.

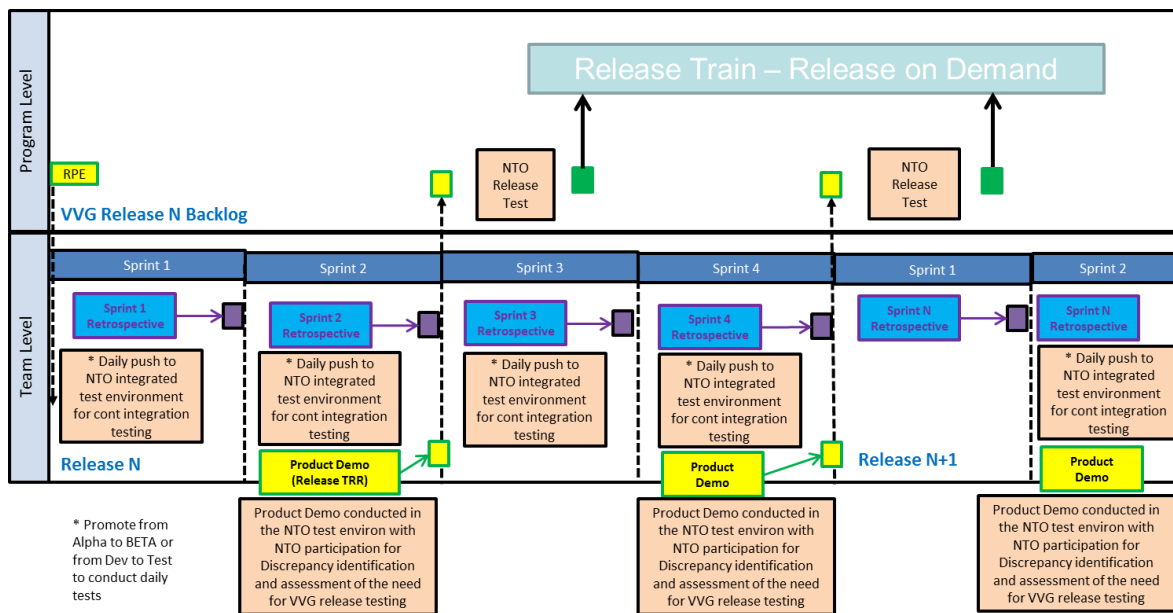
3.4.3.3 (U) Agile Test

(U) The Agile Test subservice provides full-service testing and analysis of entities developed using an agile methodology. Using current software development practices, such as agile, software is continuously tested throughout the iteration or sprint cadence. Testing is a phased approach that occurs at the Define, Build, Test (DBT) Team (also known as the development team) level, as well as the IV&V government acceptance testing completed by TAC. Figure 3 shows a general overview of how code progresses through the development cycle, into internal testing, before being promoted to formal testing in the Beta environment, and after successful testing at the NTO into the NGA operational environment.



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Figure 3. (U) Test Overview

(U) The IV&V team tests a completed feature, which is made up of completed user stories. The team verifies that the feature meets the definition of done (defined by the PO and stakeholders at the RPE) and validates the integration and interoperability of the feature within the enterprise. NTO and NIQU can participate in testing at any time in the release as long as there is a completed feature or product is ready for testing, as depicted in Figure 4.



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Figure 4. (U) NTO and NIQU Agile Test Engagement

(U) The release testing conducted at the end of a release is the final validation that the delivered feature provides the expected overarching capabilities as defined in the RPE, has been effectively integrated into the testing environment, and satisfies the acceptance criteria and nonfunctional requirements. Based on the IV&V results, TAC issues a “go” or “no-go” recommendation for transition to operations.

3.4.3.4 (U) Web Services Test

(U) The Web Services Test subservice provides testing of new and modified web services used by NSG/ASG systems. Using test automation tools, service functionality is exercised by executing scripted requests to produce known service responses.

3.4.3.5 (U) Regression Test

(U) The Regression Test subservice provides testing and analysis of entities that have been deployed to operations and have software changes (e.g., DR fixes) that require verification. These tests are termed “regression” or “enterprise regression” tests, depending on their scope.

(U) After new products have been deployed to operations and throughout the sustainment phase of the system’s lifecycle, new software maintenance releases, changes, updates, and patches will be developed to address entity DRs or other problems. TAC tests these software changes with the applicable operational architecture(s) to ensure that they correct the intended problems and that no unintended impacts to the enterprise have occurred as a result of introducing the change. When appropriate, JITC provides a Testing Status Letter to waive

interoperability testing.

(U) A regression test may be deliberately planned and conducted as a separate test event, or it may be accomplished by executing the necessary test procedures during another test event. All components of the architecture may be required to support regression testing when any component of that architecture has made a change. The TD will coordinate with affected stakeholders concerning the required level of participation for each regression test.

(U//FOUO) RTM is a type of regression test conducted specifically with the NRO. RTM testing is similar to initialization testing but is normally tailored to focus on the issues that caused the particular sensor or spacecraft to be taken offline. RTM testing is done in close cooperation with the mission partner and is reported to appropriate levels within NGA.

3.4.3.6 (U) Emergency Test

(U) The Emergency Test subservice provides support for short-notice tests that are executed to verify correction of high-priority (i.e., Priority 1 or 2) DRs (including degradation of system performance) that were discovered in operations. Emergency tests are accorded special priority for scheduling to permit the fix to be delivered into operations with minimal delay.

(U) The TDs coordinate the scheduling of test activities to verify high-priority DR fixes, whether the DR was recorded during an ongoing test activity or was discovered by operational users and recorded against the approved product baseline. The DR fixes will be verified as soon as possible to enable developers to maintain their development schedules or to permit the deployment of the correction into operations with minimal delay. For lower priority DRs, the TD will determine if the DR correction can be verified during current test activities or should be planned as part of a future test activity.

3.4.3.7 (U) Information Assurance Vulnerability Alert Test

(U) The IAVA Test subservice provides an impact assessment of the security-driven software and configuration changes to the Web browsers and JAVA Runtime Environment (JRE) used by NGA. It includes testing for IAVAs, information assurance vulnerability bulletins (IAVBs), and information assurance vulnerability technical advisories (IAVTs).

3.4.3.8 (U) Operations Assessment

(U) The Operations Assessment subservice provides informal testing of interfaces, applications, and data using ad hoc manual procedures and/or automated testing on an operational network.

3.4.3.9 (U) Development and Operations Test

(U) DevOps is the practice of development and operational engineers working together during the entire service lifecycle. It is dependent upon efficient and effective conveyance of information to and within a development team. TAC

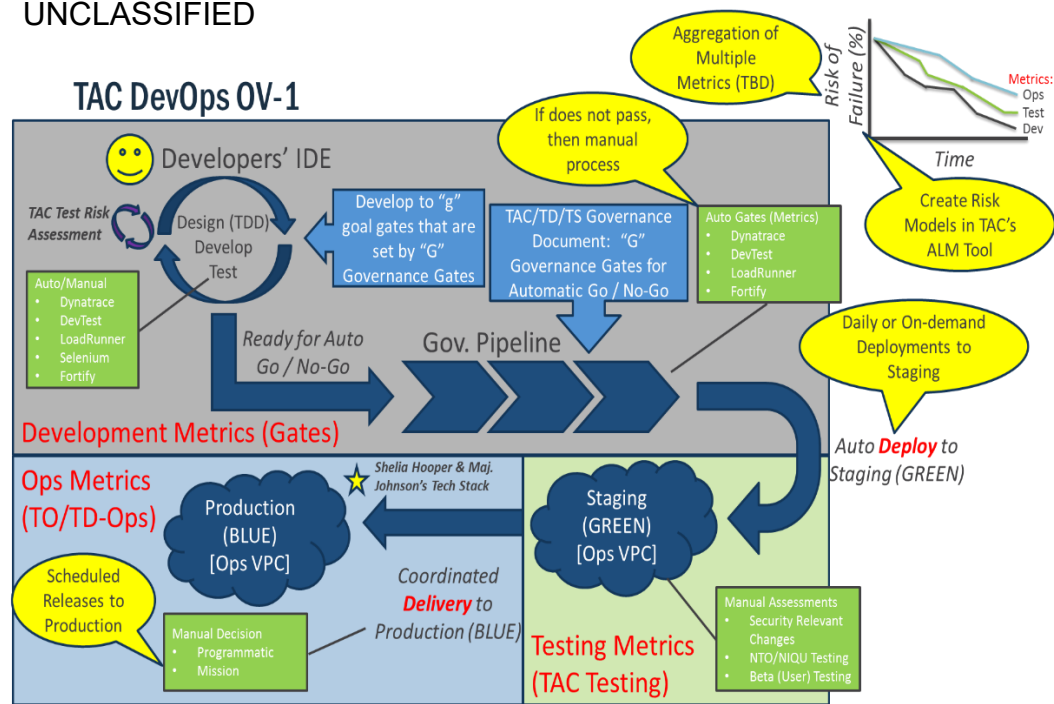
team members support the networks and hardware that support TAC's cloud test environment and the design of workflows within NGA to enable the process. This is accomplished by collaborating with all appropriate stakeholders, inserting tools for automated testing, implementing standards and guidelines when applicable, and testing finished capabilities.

(U) The DevOps process allows QA to satisfy the customer through early participation in the design and continuous assessment of delivered capabilities. It facilitates the support of changing requirements even when late in the development cycle. The intent is to deliver working functionality frequently, within days, weeks, or a few months. Working software successfully delivered by sound processes is the primary measure of progress.

(U) The developers are responsible for managing software through the DevOps workflow. A developer commits a code change to the source control repository. The build management software detects this code change and executes the deployment management tool to build, test, and package the code. The deployment management tool runs automated testing, code quality, security, and performance checks. All developers are responsible for ensuring that their codes pass these steps. The built code then gets fully packaged by the deployment management tool, with all of its dependencies so that the working software can be installed on any environment. Once the software is packaged, the generated artifacts get archived in the release management repository, depending on the development team's release cadence. Software can then be deployed to a staging environment, for independent integration testing to occur on the software before releasing it to a production environment. Other tools exist that allow the development team to automate the configuration of environments, deployment of releases to environments, and integration technology infrastructure.

(U) TAC's expectation is that developers provide smaller capability units that will need to be integrated with the NSG/ASG to conduct a user workflow of some kind. TAC will conduct automated testing to identify higher areas of risk upon which to target exploratory testing, if needed. Risk-based testing philosophy will be applied to capabilities delivered by a DevOps pipeline, just as is done with IV&V and agile testing. Figure 5 depicts the TAC ConOps for DevOps engagement:

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Figure 5. (U) TAC DevOps Testing Operational View

3.4.4 (U) Readiness

(U) The Readiness Office facilitates the NGA readiness process to ensure that milestones as defined by a program's development plan are accomplished and documented. The readiness process includes governance for major systems acquisitions, agile development, DevOps, and organizational change assessments. The processes for Readiness are established in the Tier 2 document.

(U) The Organizational Management Review (OMR) board comprises government office representatives from each CIO-T key component. The OMR assesses the effectiveness of organizational change and provides recommendations.

(U) The Readiness Office facilitates the following services:

- Establishment of the Readiness Review Team
- Determination of readiness review type
- Facilitation of required readiness reviews
- Identification of the Milestone Decision Authority (MDA)
- Preparation and distribution of the Milestone Decision Memorandum

3.4.5 (U) Security Assessments

(U) The Security Assessments service is an organizational security process that is performed as part of IV&V and regression testing by the Software Assurance Team. This process is in place to verify that security requirements are met in accordance with ICD 503 using the controls published in National Institute of Standards and Technology Special Publication (NIST-SP) 800-53r4, "Security and Privacy Controls for Federal Information Systems and Organizations." This service also verifies that system vulnerabilities have been addressed as required by the United States Cyber Command (USCYBERCOM).

(U) The Defense Information Systems Agency's (DISA's) Assured Compliance Assessment Tool (ACAS) is the NGA security assessment tool of choice and is run on each part of the delivered configuration to identify any relevant security issues.

3.4.5.1 (U) NGA Software Assurance Process

The Software Assurance Process (SWAP) aligns with the ICD 503 Risk Management Framework (RMF) by fulfilling security review and enforcement requirements concerning the integrity, security, and reliability of all open-source, freeware, shareware, COTS, GOTS, and foreign software used by NGA programs. The SWAP Team executes a disciplined, robust, and repeatable software evaluation process and fulfills the following in support of ensuring critical quality parameters are met:

- Providing the research and investigation required to identify if known vulnerabilities exist in software, either maliciously or unintentionally inserted (accomplished through Common Vulnerabilities and Exposures (CVE) and open-source collection).
- Providing the body of evidence to the Designated Authorizing Official, allowing for agency software risk approval or disapproval.
- Providing the analysis reports to the requesting programs that outlines all software vulnerabilities for remediation.
- Coordinating with the NGA Security and Installation Operations (SI) Directorate on the review of foreign software
- Categorizing software based on potential risk indicators
- Assuring alignment with ICD 503 RMF
- Performing static code analysis of all custom applications and software with accessible code
 - Providing analysis of scanned code to developers for remediation of findings
 - Coordinating the deployment of a designated static code analysis Tool within NGA development environments in concert with CIO-T

Directorate

3.4.6 (U) Business Process Management Team

(U) TAC's Business Process Management (BPM) Team works with NTO and NIQU to identify opportunities for process improvement; the team initiates and develops recommendations for process solutions. The BPM Team provides technical editing services and also conducts QA and security reviews of NTO's and NIQU's test documentation to ensure TAC delivers quality and accurate products. The team supports development of TAC process assets (e.g., templates, checklists) and provides process and QA training to TAC personnel.

(U) The BPM Team works with TAC management to identify metrics that will drive business decisions. The team collects, analyzes, and reports on the metrics as directed by TAC management. The team also identifies and shares trends with division chiefs to indicate areas of progress or areas that require attention and change.

4 (U) NGA TEST ORGANIZATION DIVISION

4.1 (U) THE NTO MISSION AND RESPONSIBILITY

(U) NGA's NTO is responsible for test planning and execution of systems and software within the NSG/ASG Enterprise Architecture. The NTO accomplishes this protection of the NSG's/ASG's operational baseline and its users from the deployment of defective applications and systems by:

- (U) Verifying that new or changed elements of the NSG/ASG Enterprise are successfully integrated into the operational baseline
- (U) Validating the inter-system functionality and fitness for use of new or changed elements of the NSG/ASG Enterprise
- (U) Providing information to support DoDIIS CtFs for NGA systems

4.1.1 (U) Certification to Field

(U) Deployment of any new development or modification to an existing entity to any NGA network or to an NGA customer site, whether internal or external, requires approval of D/TAC. Based on the results of testing, the NTO prepares a Deployment Recommendation/Decision Summary for D/TAC signature. When this document is signed, the NTO publishes a CtF announcement. This process is used for all NGA products, regardless of whether they are deployed to DoDIIS or non-DoDIIS sites.

4.1.2 (U) Waivers

(U) When warranted, any PM can request a waiver from D/TAC for part or all of the independent testing activities.

4.2 (U) NTO STAFFING

(U) The NTO is staffed by an integrated team of government and contractor test engineers (supported by a dedicated contractor team of process management) and application services support personnel.

(U) Provisioning of mission systems hardware, software, and associated support staff (field service representatives) for the NTO TE is provided by the PM as part of the program budget. Segment test hardware and software at the NTO TE is maintained as part of the program recap schedule, and each PMO provides dedicated staff (field service representatives) as needed to operate its systems in the test environment. With the implementation of the Application Service Provider/Infrastructure Service Provider (ASP/ISP) construct, this responsibility is now split between the ASP and the ISP in consonance with the separation of responsibility for application and infrastructure portions of the program or project.

(U) Each NGA segment has individualized requirements relating to cross domain workflows. To meet this need, capabilities may be delivered to several

operational domains to satisfy mission intent. The NTO TE supports capabilities delivered to NGA and IC agency partner datacenters residing across Sensitive But Unclassified (SBU), Secret Network (SECNet), and Joint Worldwide Intelligence Communications System (JWICS) networks as well as IC cloud ISPs, including Amazon Web Services (AWS) on the Unclassified Cloud (UC), Secret Cloud (SC), and Top Secret Cloud (TC).

4.3 (U) NTO QUALITY CRITERIA

(U) The NTO requires a successful TRR to enter into IV&V testing or regression testing. The NTO success criteria for all testing activities are based on the priority level of the found discrepancies and on the cumulative number of issues and discrepancies on any one or more functional areas. TAC's criteria for the priority level of the discrepancies are described in NGA Configuration Management Plan (NCMP) Rev D and in Table 1.

Table 1. (U) Discrepancy Report Priority Definitions Used in Testing

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Priority Code	Definition	Response Time (in Operations)
1	Major in line process does not function. Operations and/or test activities cannot continue.	Fix immediately or develop a workaround within 24 hours.
2	Major process/functionality significantly degraded. Normal operations and/or test activities severely impacted.	Fix or develop a workaround within three days.
3	Ancillary process does not function or Priority 1 or 2 discrepancy is being downgraded because a viable workaround exists.	Investigate, fix, and/or provide status within 14 days.
4	Minor impact to operations and/or test activities and/or enhancement to capabilities relating to routine operational improvements that have no direct operational impact.	Investigate and fix when possible considering other priority tasking.
5	Documentation updates that have no direct operational impact and for TBR/TBD resolutions unless the resolution warrants a higher priority.	Update the documentation when possible.
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4.4 (U) NTO TESTING FACILITIES

4.4.1 (U) NTO Test Infrastructure

(U) The NTO manages a test environment that is separate from, but that closely

represents, the operational environment, whether that environment is in the cloud or in a data center. Entities that run in a virtual environment in operations are tested in a virtual environment within the NTO TE; entities that run on “bare metal” servers in operations are tested on bare metal servers within the NTO TE. This high-fidelity test environment permits NTO test personnel to validate the successful integration and functioning of the entities under test without interfering with or posing a risk to the actual operational baseline.

(U) The NTO test environment supports testing of both NSG/ASG Enterprise mission systems (e.g., Analytical Services Contract (ASC)/Stellar, GEOINT Visualization Services (GVS), GIMS, and Cedalion) as well as corporate networks (e.g., SBU, SECNet, Common Operating Environment/NGA Network (COE/NGANet)). Testing of NSG/ASG Enterprise mission systems is conducted primarily at the NTO TE in the NCE, while corporate network and desktop testing is primarily performed at the NTO TE in the NCW. Each of these locations has a limited capability to provide backup support to the other’s mission.

(U) Planning for the maintenance and evolution of this independent test environment must take into account the following criteria:

- (U) NTO CM control of the NTO TE and isolation of the NTO TE from all other activity, including developer testing and system changes, are required to ensure testing validity and integrity.
- (U) Test capacity required for the virtualized environment must be determined with the assistance of ISP analysis of current test loading and projected growth for what is to be virtualized.

4.4.1.1 (U) Hardware and Support Staff

(U) The existing hardware provisioning method will continue to be used for products that have been delivered and supported by ASP providers in the past. The ASP must continue to provide factory service representatives (FSRs)/engineers to support testing of their entities. FSRs/engineers must have specialized application expertise beyond the capability of the current ISP organization in order to provide the necessary level of participation in NTO testing.

(U) For virtualized products and other products for which the ISP provisions, operates, and maintains the Information Technology infrastructure, the ISP will provide the same support for the test environment. Reliance on as-available use of operational information technology support personnel to support test systems tends to degrade the responsiveness, flexibility, and test capacity that the NTO is capable of providing to NGA. To meet the required level of NTO responsiveness, staff support to test systems are rapidly available.

(U) For widely deployed segments requiring testing in multiple architectures, the NTO must possess enough hardware to test multiple configurations at the same time in order to provide the minimum acceptable level of responsiveness to NGA

and mission partner transition schedules. It has been the NTO's experience that at least two ITs are needed prior to an IV&V for a major build for legacy systems, with at least a week of DR work-off time after each. Hardware assessments need to take this into account when determining the right amount of equipment to provide to the NTO for testing. Although virtualization will greatly increase the efficiency of hardware utilization, the need to test multiple versions of entities will continue to apply for virtualized applications and services just as it does for legacy systems. Hardware allocations for testing of virtualized and non-virtualized applications and services must take into account the requirement to run multiple copies of key entities in separate test "strings" simultaneously. These test strings must be configured to represent the configuration in operations as closely as possible.

4.4.1.1.1 (U) Software Versions Support

(U) A deployed entity's software in operational use must be maintained and, in many cases, may be upgraded. During its operational lifetime, one application or service can provide a series of two distinct types of software upgrades. The first type is a routine corrective Operations and Sustainment (O&S) patch to the currently fielded version of a segment or service, which requires regression testing to ensure the patch fixes the targeted discrepancies without introducing new problems into the operational enterprise. The second type is the next version of that segment or service that adds new functionality and/or incorporates new COTS releases, which may require ITs and an IV&V.

(U) For virtualized systems, the same need exists for a sufficient safeguarded and guaranteed capacity to allow for simultaneous testing of multiple segment/service versions.

4.4.1.1.2 (U) Emulation of Operational System Capacity

(U) The ability to do full performance testing is heavily dependent on the degree to which the test information technology infrastructure represents the operational infrastructure. Given the size of the NSG/ASG Enterprise, provision of a full replica of the operational information technology infrastructure is not realistic. However, considerable risk reduction is achieved by providing a representation of the operational environment that uses the same hardware and software where feasible and relies on representative systems, such as scaled-down versions and test partitions, for the larger and costlier items. The use of virtualized environments and cloud infrastructure should improve TAC's ability to provide more accurate performance testing results. The objective is to determine and implement a test environment that achieves the right balance between cost and benefit in terms of realism versus risk.

4.4.1.2 (U) Test Workspace

(U) The NTO test workspace at NCE and NCW comprises test cells. It is in these cells that the majority of NTO testing is accomplished, although some tests, such

as those requiring only a single tester, can be performed from the tester's desk. These cells contain production-representative workstations connected to the test versions of the various NSG/ASG systems established in the NTO TE. Each test cell can be configured as needed to represent typical user workflows.

(U) The FSRs/engineers and ISP engineers support testing from their desks; their cubicles, desks, and workstations are located in a large centralized work area adjacent to the test cells as well as virtually. The collocated work areas facilitate coordination and are essential to successful test execution.

4.4.2 (U) Engineering Test Data/Simulated Reference Data

(U) In the interest of improving the realism of tests, most NTO and NIQU tests are accomplished using transactional-based data that is generated by actually operating the systems during tests and through a process of ripping wideband operational data from operations (e.g., ripper). However, in those circumstances when the ripped operational data and the generated data are not available, the availability of adequate engineering data (e.g., simulated wideband and narrowband system data) is critical to the success of testing.

(U) Engineering test data is used to support NTO independent test activities as required, including ITs, IV&V, and regression testing. The TD, working through the test IPT, shall identify the need for engineering data to support end-to-end test activities. Engineering data will be needed to represent exchanges and transfers among and/or between NGA systems or products and external elements. The test data shall conform in format to the actual data being simulated in terms of file size, data structure, and valid data values. An important criterion in defining requirements for engineering data is the degree of correlation needed across data fields and across separate data files and messages. For wideband requirements, it is important to identify the specific type, number, and quality of images needed. In some situations, the NGA Enterprise Scenario Generator (NESG) may be used as a scenario/test case driver to supply correlated test data as input to the tests. In other situations, direct manual intervention by the segments will be required to create test data that is correlated to the reference data. Acquisition of engineering data to support NTO and NIQU testing is the responsibility of the PM.

4.5 (U) NTO CUSTOMERS

(U//FOUO) The NTO serves many customers as part of its enterprise QA mission. The NTO is involved in every step of NGA's acquisition process, from concept development through transition to operations, as noted below:

- 1) Internal NGA Research and Development (R&D) offices draw upon NTO's operational experience to help evaluate systems against an operational user workflow in order to understand better those systems, services, and products intended for technology insertion.

- 2) NGA's PMOs, POs, and capability owners use the NTO as a key risk mitigation to their deployment decisions through identifying, resolving as appropriate, and validating DR fixes for system and service discrepancies in order to ensure enterprise quality capabilities.
- 3) NGA's user community (i.e., United States Combatant Commands (US COCOMs), NGA Source, NGA analysts, and military services) look to the NTO as an enterprise QA organization that ensures that the systems delivered to them for mission capability operate as needed.
- 4) Multiple mission partners (e.g., NRO, commercial data providers (CDPs), airborne system program offices) look to NTO as their partners in the joint enterprise QA process, which is accomplished through enterprise testing.

4.6 (U) REQUESTING NTO SUPPORT

(U) The NTO RFS Tool is a web-based software package that integrates the NTO's core workflow and supporting management functionality into one common logical framework. It is used by the requester of the test, the NTO TD, members of the NTO test support staff, and the ISP to coordinate and schedule the full range of test and support resources needed for the test. The tool is accessible from COE, NGANet, and JWICS and provides a virtual door into a series of custom software applications that meet a variety of NTO management and technical needs.

4.7 (U) NTO TEST PROCESS

(U) Section 3 of the NSG/ASG Enterprise TEMP articulates the TAC's integrated test program in terms of the various types of testing conducted throughout the lifecycle of NSG/ASG entities. This section provides a general description of the activities, forums, and documentation used by the NTO to plan, coordinate, execute, and report on the tests for which it is responsible. It should be noted that this is not intended to be an exhaustive treatment of internal NTO processes and that documentation formats are tailorable to fit the specific needs of each program and project.

(U) This section presents NTO test operations as being a sequence of clearly delineated activities. Planning and coordination necessarily precede execution which, in turn, must precede reporting of results. In practice, it is not uncommon for these activities to overlap, either to compensate for limited test resources or to expedite entity development schedules. Furthermore, various agile development and other innovative acquisition initiatives (such as DevOps) require the NTO to tailor these activities to accommodate their program-unique requirements. In the interest of brevity and clarity, this section will not address the specifics of these variations. Only a conceptual, idealized process sequence is presented.

(U) Although the NTO TD does serve as the facilitator and POC for the PMO on all IT and product demonstration RFSs, this section does not address those

actions because ITs and the product demonstrations are a PM/PO-led activity and are frequently run by the developing contractor in coordination with the PM/PO. These activities can use the NTO TE and are encouraged to do so, but there is no requirement for those activities to be executed at the NTO TE or with the systems, segments, or equipment in the NTO TE.

4.7.1 (U) Test Planning and Coordination

(U) NTO test planning begins early in the system lifecycle with designation of an NTO TD who participates in milestone events and other planning forums. The TD reviews system documentation, attends requirements and design reviews, participates in release planning events, and observes developer FATs or demonstrations to understand how the system operates and to establish sound NTO test objectives. The TD also ensures that the program schedule contains adequate provisions for testing and coordinates test windows within the NTO schedule. The TD participates in all applicable readiness events as a member of the Readiness Review Team and briefs the NTO assessment of readiness for NTO testing at any TRRs that are traditionally the control gate to NTO testing.

(U) As early as feasible in a program's development, the PMO and PO—in consultation with the NTO TD—will prepare and submit an NTO RFS. This submission reserves an NTO test window and alerts the NTO test elements that they can expect to be involved in this test. The RFS can be submitted even if all information is not yet available. The projected test will be entered into the NTO resource allocation system through the Resource Allocation Meeting (RAM) and gets added to the NTO's Master Schedule. For any RFS that is RFC-driven, the RFC number should be recorded as soon as it is available.

(U) Each program schedule is unique; the NTO process is tailorable to the needs of the program. The objective is to provide responsive support to the program, allowing it to adhere to its schedule while continuing to provide an acceptable level of risk management.

(U) Planned system deliveries normally allow for test preparations to proceed deliberately in alignment with the readiness process. However, the NTO is routinely required to insert unplanned emergency tests into its schedule or to re-plan a test to help a program to recover a deployment schedule jeopardized by delays in development or test failure. In these cases, coordination becomes more challenging. Although the NTO will make every effort to accommodate short-notice support requests and changes, such accommodations usually require the support of contractors who maintain portions of the NTO TE but are not under the NTO's direct control. The NTO, in coordination with the respective PMOs, actively works with external contractors to ensure the necessary responsiveness is provided to support the NTO's mission and the program's schedule needs.

(U) Based on the RFS and the TD's knowledge of the development, the TD will prepare a test plan or pre-test risk assessment for the event. Additionally, the TD

will prepare test objectives/features and procedures based on the PMO's or PO's operational scenarios or workflows.

(U) Approximately one or two weeks before the test begins, the NTO will hold an NTO Internal Test Readiness Review (ITRR). This is a final check on the readiness of the NTO to execute the event. The TD submits a completed ITRR checklist (found in the RFS) during the meeting that is designed to ensure that late-breaking changes have been accounted for and that visibility is being maintained on any unresolved issues or action items.

4.7.1.1 (U) Test Integrated Product Team

(U) Most NTO test planning and coordination, while thorough and rigorous, is conducted informally by the TD who interacts with the test event stakeholders through ad hoc meetings or one-on-one contact. However, when the NTO TD determines that a test is too large or complex to be planned and coordinated in this fashion, the NTO TD will establish a test IPT to facilitate the collaboration of all test event stakeholders to ensure the successful accomplishment of these tasks. The TD chairs the IPT whose members include representatives from the NGA PM/PO, development contractor, TSR, Software Assurance, NIQU, JITC, NTO test support staff, the ISP (when applicable), other NGA internal stakeholders, members of the user community for this capability, and the external stakeholders participating in the test activity. The Test IPT provides the following:

- Communication among the participants
- Adding structure to the test planning
- Coordinating a process to ensure a common understanding of test requirements and scope

(U) All required resources for the test are identified, planning timelines are met, and changes are promptly identified and accounted for.

4.7.1.2 (U) NTO Test Planning Meeting

(U) The NTO Test Planning Meeting (NTPM) is a DoDIIS certification event (though it is sometimes convened for non-DoDIIS developments as well) that is chaired and coordinated by a NIIST member. An NTPM is held in advance of a major system release or when, in the judgment of the NIIST Team Chief, it is necessary to ensure the successful planning and execution of the DoDIIS certification process for any specific delivery. The participants are the NTO TD, PM/PO representative, the developer, JITC representative, and Software Assurance representative. All parties' requirements and concerns should be discussed and fully understood. The NTPM chair ensures that all parties have a clear understanding of the DoDIIS certification process, test planning, deployment planning, and the roles of each organization.

(U) In advance of the meeting, the PM/PO will provide details on the following aspects of the test event:

- (U) Verification that all required documentation was provided on time
- (U) Type of release (major, minor, maintenance)
- (U) NTO test plan review (scope of testing, test dates, deployment schedules, POCs, etc.)
- (U) JITC test plan review (scope of testing, test dates, list of interfaces, POCs, etc.)
- (U) Security testing (scope of testing, test dates, test site, POCs, etc.)
- (U) Verification that all required hardware and software are available
- (U) Verification of successful DT&E
- (U) Listing of FAT findings
- (U) Listing of any open CRs, problem reports (PRs), and DRs that may exist against the baseline system.
- (U) Detailed schedule of test activities

4.7.1.3 (U) Test Objectives/Features

(U) The NTO TDs will use available entity documentation in conjunction with knowledge gained through participation in milestone events and other interactions with the program office and developer to formulate applicable test objectives for NTO test events. The documents typically used for this purpose include CONOPS, the NSG Requirements for Application and Infrastructure (NRAI) database, Capability Development Documents (CDDs), system specifications, and JIRA contents (e.g., features, epics). The test objectives/features guide the verification and validation of the entity's successful integration and functioning with the systems with which it is required to interoperate. Most objectives/features are formulated to be achievable using a DITL test approach that exercises the entity in the NTO TE where functional threads, chains of events, or business processes are executed as they would be in typical real-world, day-to-day operations. However, if circumstances warrant (e.g., there is a need to verify a requirement that could not be verified during FAT), objectives/features may be developed that require the NTO to use alternative test approaches. More details on the creation of objectives and features can be found in the TD handbook.

4.7.1.4 (U) Test Plans

(U) The TD will normally create a separate test plan or pre-test risk assessment for each formal test event (IV&V and regression testing). The TD will create and provide to the PMO/PO a test plan for those entities that are being tested for the

first time. The TD will create and provide to the PMO/PO a pre-test risk assessment document for any entity that is providing an updated build and/or fixing discrepancies. When multiple effectivities or software releases are tested together, the TD will have the ability to create a consolidated test plan. In addition, test plans are written for joint interoperability test activities supported by JITC personnel collocated with the NTO. More details on the creation of test plans and pre-test risk assessments can be found in the TD handbook.

4.7.2 (U) Test Execution

(U) The NTO tests operational user community workflows within the NSG/ASG mission systems architectures. These tests are executed by an integrated cadre of government and contractor systems engineers who are trained to operate the NSG/ASG systems as the users do. They use the NTO TE to exercise the end-to-end data flows of the NSG/ASG Enterprise, primarily using DITL activities to identify any interface problems. They are directly supported in test execution by the FSRs/engineers for the participating segments. Some provide direct, specialized support to test execution by setting up the imagery workflow that queues imagery for the test engineer's account, much as they do in operations.

(U//FOUO) Effective test execution relies heavily on the synergy of the TD and FSR/engineers. FSRs/engineers at non-collocated sites, such as those managing the NSG Consolidated Library (NCL) or GEOINT Data Transformation Service (GDTS), participate in secure teleconferences held as needed throughout the test day. Each test suite is equipped with both secure and unclassified Polycoms to facilitate teamwork and coordination. During joint tests with the NRO and other major tests involving multiple sites, a secure voice telephone number will remain active throughout the test day.

(U) The TD chairs daily start-up and wrap-up teleconferences for test participants and stakeholders. These meetings are used for discussing current technical issues, sharing information, and updating system and test status with interfacing segments. Frequently, the TD and PMO will hold separate engineering review boards (ERBs), after the wrap-up meeting, to discuss in detail the discrepancies found during the test day that are specific to the segment under test.

(U) The PM/PO and system integrator representatives are kept informed of test progress and issues continually throughout the test day. They are included in the daily start-up and wrap-up meetings and receive daily status reports via email. When the seriousness of the issue dictates, they are directly contacted by the TD and included in problem resolution meetings on the test telephone number.

4.7.3 (U) Test Reporting

(U) At the conclusion of a test, the NTO will prepare the following reports:

- (U) The TD will prepare a final status report for the functional/integration portion of the test

- (U) The NIIST will prepare a report for the installation audit portion of the test
- (U) The NIIST will prepare a report for the security scan portion of the test
- (U) The JITC representative to the NTO will prepare a report on JITC support of the test (for DoDIIS systems only)

(U) The PMO/PO can expect that all reports listed above will be available to them within two weeks of completion of the event and that the reports will document significant activities that occurred, including issues encountered and any resultant DRs.

(U) Upon receiving approval of the security report from the reviewing Risk Management representative, the NIIST will prepare a CtF package for approval by D/TAC. The contents of this package depend on whether the tested entity was a DoDIIS or non-DoDIIS product and, if the latter, whether a Beta II test was required (more details in Section 3.4.3.1).

(U) Additional details of the final test report and its timelines can be found in the TD handbook.

4.8 (U) NTO TEST SUPPORT FUNCTIONS

4.8.1 (U) Test Information Technology Infrastructure Operations

(U) TAC systems administrators (TAC-SAs) and FSRs/engineers provide IC&T as well as ongoing Operations & Maintenance (O&M) of NGA information processing infrastructure and NSG/ASG production systems under test. More specifically, TAC-SAs and FSRs provide O&M and troubleshooting support to TAC systems in the physical, virtual, and cloud computing domains. As SMEs for their respective areas of support, TAC-SAs and FSRs coordinate daily with NTO test engineers and TDs to facilitate testing schedules, new capability training, and communication with factory-based developers and PMOs.

4.8.2 (U) NTO DR Management

(U) The NTO Problem Incident Management (PIM) works closely with the NTO test engineers during testing, providing their expertise on the NGA DR tracking system (Remedy) to support reporting and tracking of DRs. PIM records key items of information about each DR in designated Remedy fields reserved for NTO use and the DR number in Hewlett Packard Application Lifecycle Management (HP ALM). Upon test completion, PIM transfers open DRs to the CS ERB for adjudication and represents NTO's interest during CS ERB meetings.

4.8.3 (U) NTO Business Area Lead Document Quality Review

(U) The NTO BAL document quality review (DQR) is the process by which BALs review and approve all NTO documents directly associated with IV&V and

regression testing that are distributed externally (e.g., test plans, pre-test risk assessments, final status reports). Additional details about the BAL DQR process are in the TD handbook.

4.8.4 (U) NTO Gold Copy

(U) The purpose of Gold Copy is to maintain control of software versions under test by the NTO and to archive verified and validated product baselines in the NTO's Gold Copy Library at the NCE or on VOLT for those entities that deliver software to production virtually. TAC CM catalogues and maintains copies of test support materials and engineering data (e.g., simulated imagery datasets) used for NTO testing.

(U) Three software baselines are used in the NTO TE. The integration test baseline is the initial version of new or upgraded software delivered to the NTO TE by a PMO where it undergoes DT&E Phase 2 (i.e., IC&T and one or more ITs) under the control of the PMO and developer. After completion of the last IT, this software is locked down by the NTO and re-designated as the operational test baseline; this is the version that undergoes government testing/OT&E. Upon successful completion of government testing and CtF, the operational test baseline is re-designated as the product baseline, which is the version authorized to be transitioned into operations.

(U) NGA PMOs provide TAC CM with the following:

- (U) Two copies of the entity's integration test baseline, if needed for installation. Those entities delivering their software baselines virtually to a data center or cloud infrastructure would not need to deliver their baselines to the NTO. For those that are delivering a copy of their baselines to the NTO, one copy will be maintained in the Gold Copy Library and the second copy will be used to check out to the NTO test staff for testing.
- (U) A list of all COTS/GOTS products delivered to the NTO.
- (U) Two copies of any test data required for testing in the NTO TE.

(U) The Gold Copy process starts when copies of the IT baseline are delivered to the NTO by the PM/PO. The NTO tracks this software through its progression to operational test baseline and—pending successful testing—its final promotion to product baseline. It is this approved product baseline that is referred to as the Gold Copy and is archived by the NTO. The developing PMO also maintains a copy of this product baseline, which it uses to affect the actual transition of the new software into operations and to meet its responsibility for restoring the software at operational sites in the event a site's version becomes corrupted. The NTO's Gold Copy is used to configure the test environment baseline for subsequent testing of other systems undergoing NTO testing. It also serves as a back-up to the development contractor's record copy in the event the record copy

is corrupted. Each new version of a product baseline will be stored as a separate Gold Copy version in the Gold Copy Library at the NCE. For those entities not delivering their baselines on media to the NTO, the list of COTS/GOTS that make up the delivery will be recorded in VOLT as a record of what the NTO tested and what the NTO recommended for deployment.

(U) Changes to approved Gold Copy product baselines will not be made until the software modifications/updates have gone through the IV&V/regression test process. All changes to product baselines—whether major, minor, or maintenance—must go through the NTO test process.

4.8.5 (U) Test Automation

(U) The test lifecycle phases include test planning, test execution, defect management, and test analysis and reporting. The TAC Test Automation Team supports these test lifecycle phases through the configuration and use of a collection of automated test management and test execution tools.

- (U) HP ALM is a web-based quality management tool for organizing and managing test artifacts, scheduling and executing test sets, collecting and analyzing test results, tracking and monitoring defects, and automating test document generation.
 - ALM is the single repository of all test objectives, test procedures, test steps and resources that are standardized, controlled and shareable across segments.
 - For test documentation, ALM auto-populates data-driven tables, minimizing formatting/template issues and outdated data.
 - During test execution, status is recorded electronically as testers perform an ordered list of test procedures (manual/automated) stored in ALM.
 - DRs are initiated and managed in ALM and tracked in both Remedy and ALM. ALM automatically populates the DR description with the detailed step in the failed test procedure.
 - ALM provides out-of-the-box and customizable real-time metrics and graphs that can aid the TD in the execution/management of test activities (DRs, objective completion, and resource allocation) and provide status to TAC management.
- (U) HP Unified Functional Test (UFT) provides two capabilities:
 - QuickTest Professional (QTP) automates repetitive manual test procedures by developing reusable test scripts. It is useful primarily for regression tests.
 - Service Test automates functional testing of services without the need for a graphical user interface (GUI).

- (U) HP LoadRunner (LR) reduces the cost and time required to manually run load, performance and application stress tests by automating and simulating multiple users.
- (U) Computer Associates (CA) DevTest Solutions provides a suite of tools that aid automated functional and performance testing of service components in a net-centric environment, and emulates highly-constrained and unavailable live services and shared systems.
- (U) Compuware Dynatrace monitors and profiles the full transaction from the user interaction to the line of code on the back-end servers to aid in identifying performance bottlenecks.

5 (U) NGA IMAGE QUALITY AND UTILITY DIVISION

(U) NGA's NIQU division is responsible for evaluating the quality and utility of all GEOINT sensor data products throughout the program lifecycle, through participation in both the development lifecycle of NSG/ASG entities and the operations cycle of day-to-day exploitation. This provides NGA confidence in the acquisition and provides NSG/ASG users with confidence in the quality of the data used to support their missions. NIQU performs this mission through the following:

- a. (U) Development and maintenance of PQ standards
- b. (U) Review of system-level PQ requirements and input to support NIQU testing
- c. (U) Support to PQ design trade studies
- d. (U) Development and maintenance of an NGA enterprise PQ testing process
 - (U) Validating storage and sensor data transformation tools
 - (U) Validating exploitation tools
- e. (U) Support the operational readiness process for transitions, upgrades, and initializations via PQ testing
 - (U) Verifying and validating current and future data sources integrated into the NSG/ASG
- f. (U) PQ operational monitoring of commercial, national technical means (NTM), and airborne data sources.
 - (U) Providing technical sensor data quality expertise to the NSG/ASG users
 - (U) Monitoring and testing the quality and utility of sensor data and products throughout the NSG/ASG architecture

5.1 (U) NIQU STAFFING

(U) NIQU consists of an expert team of imagery scientists, statisticians, imagery analysts (IAs), photogrammetrists, software developers, and systems engineers. The NIQU QA effort is led by the Operations Lead (OL), who is responsible for the functional and technical direction of the integrated team. The integrated team supports a wide array of technical topics and activities across the entire spectrum of the NGA enterprise. These topics and activities support the overall NIQU mission.

5.2 (U) PRODUCT QUALITY ASSESSMENT

(U) PQ across the acquisition lifecycle is verified in accordance with the performance thresholds and objectives stated in KPP 1, 3, and 4 as defined in

the NORD. For KPP 1, NIQU satisfies top-level interoperability requirements within the image chain by verifying that sensor data can be successfully exchanged between nodes in the image chain. NIQU supports the requirements for processing, exploitation, and storage by verifying that sensor data can be successfully processed and exploited. NIQU ensures the KPP 3 requirement to maintain the quality of sensor data through the image chain by verifying that sensor products suffer no loss in quality or utility. NIQU supports KPP 4 requirements to ensure end-to-end information integrity by verifying that sensor data is not corrupted as it passes through the image chain.

(U) A PQ assessment is conducted by NIQU personnel either independently or in conjunction with NTO testing. It is conducted on products produced by programs or projects, on sensor data or products provided by NGA, on mission partner assets, and on sensor data or products from commercial or airborne providers. This testing can occur at any point in the developer or government testing timeline. These activities are intended to ensure that every system that introduces sensor data into the NSG/ASG or manipulates sensor data or product data within the NSG/ASG meets and maintains nationally recognized standards for interpretability.

(U) NIQU verifies PQ-related, enterprise-level functional requirements for systems undergoing a transition to operations while verifying PQ performance. These requirements cover the range of functionality from collection and storage through dissemination and exploitation for a variety of imaging modes. Verifying PQ performance also includes conducting accuracy assessments of Image Quality Equations (IQEs). As part of most transitions, NIQU verifies that sensor data products are successfully collected, stored, and disseminated to exploitation workstations and that key functions of storage, manipulation, and exploitation operate correctly with the sensor data product type being evaluated. The functions examined include but are not limited to product chipping, product transcoding, product reformatting, product ingest, product display, smooth roam, smooth magnification and demagnification, support data compliance and accuracy, geo-location and mensuration accuracy, contrast enhancement, sharpness enhancement, and dynamic range adjustment. The results of NIQU's functional testing are reported as part of the overall NGA test reports or initialization test report. In addition, NIQU may also conduct quality evaluations on algorithms at the request of various mission partners and standards boards for potential future applicability to the NSG/ASG.

5.3 (U) PRODUCT QUALITY CRITERIA

(U) NIQU uses the PQ-ORC when assessing the performance of NSG/ASG segments and the overall imaging enterprise. The criteria are used to organize the test planning process and provide clear traceability back to PQ policy guidance and system requirements.

5.4 (U) NIQU TESTING FACILITIES

5.4.1 (U) NIQU Engineering Cell

(U) The NIQU Engineering Cell is equipped with operationally connected IEC workstations that are used to monitor operational PQ, reproduce existing operational PQ issues and conduct pre-operational checkout of sensors during initialization. The Engineering Cell is compliant with the Softcopy Exploitation Facilities Standard (SEFS).

5.4.2 (U) NIQU Evaluation Facility

(U) The purpose of the SEFS-compliant NEF is for collection of controlled image quality data and analysis of non-operational data on workstations that are compliant with the Softcopy Exploitation Display Performance Standard (SDPS). The workstations in the facility are set up specifically for conducting image quality experiments. The software used on the workstations is the Viewer for Image Processing, Evaluations, and Reviews (VIPER) software package. It is used to collect Absolute National Imagery Interpretability Rating Scale (NIIRS) ratings, Delta-NIIRS ratings, and Engineering NIIRS Standard ratings. The evaluation room is often used during initialization activities to conduct IQE accuracy evaluations. It is also frequently used for capability development by enabling controlled experiments of functions, such as compression and enhancement, that support PQ.

5.4.3 (U) NIQU Use of the NTO Test Environment

(U) NIQU uses the NTO TE to conduct PQ assessments of new or enhanced NSG/ASG entities, such as storage, manipulation, and exploitation of software and hardware and to acquire image data in support of sensor initialization activities. NIQU coordinates with the NTO for all PQ testing associated with NTO test activities at the NTO TE. The NIQU staff also participate in NTO management processes in order to better integrate PQ testing into the NTO workflow.

5.4.4 (U) NIQU Testing in the Cloud

(U) NGA is embarking upon a new paradigm, cloud computing, in an effort to increase efficiencies. This activity means that NGA will transition from standalone hardware loaded with software to scaled computing with virtual software. The benefits to NGA include:

- Scalability providing responsiveness to fluctuating bandwidth demands
- Increased automation of processing thereby increasing speed of capability
- Elimination of hardware and maintenance purchases reducing costs
- Cloud-based workflow and file sharing applications increase collaboration capability within the IC

(U) NIQU will embrace this new paradigm and insert itself in scheduled phased events. Those phases include the establishment of a cloud architecture, security

implementation, tool transition, data migration and defining assessment methodologies.

5.5 (U) REQUESTING NIQU SUPPORT

(U) NIQU participation in a program or project's development and operation is triggered by specific events and directives. In most cases, NIQU initiates PQ assurance in a program/project because a review of the RFC documentation provides evidence of a PQ risk. Changes to the sensor data handling software or dissemination architecture are examples of the types of modifications that NIQU looks for when deciding to direct its efforts on a particular program.

(U) NIQU becomes involved in NTO IV&V testing via the NTO RFS process. When a program manager uses an RFS to initiate a request for NTO testing, image quality is one of the test participants that can be selected. If the program manager does not select image quality as a participant, the NIQU representative or NTO test director may add image quality upon reviewing the nature of the entity to be tested.

(U) NIQU participation may also be initiated through NGA management directive. NIQU can receive directives for PQ analysis from each of the NGA directorates. NIQU can also receive directives for PQ analysis from organizations external to NGA, including the NRO and other data providers that often request NIQU participation in joint PQ activities. After vetting the request through D/TAC, NIQU will conduct the appropriate PQ analysis.

(U) NIQU is also introducing additional processes to accommodate the agile environments within NGA, such as DevOps and Cloud migration. These processes are currently evolving but still require the concurrence of D/TAC.

5.6 (U) NIQU TEST PROCESS

(U) NIQU is involved in the program lifecycle for pixel-processing capabilities for NGA-acquired systems and participates in all key mission partner activities.

(U) There are three stages in the typical test process employed by NIQU. Prior to FAT, NIQU often aids the development contractor and program office by providing an informal, early assessment of the format and support data compliance and accuracy. These assessments are sometimes used by the program office during the PSR to determine if the system is ready to advance to integration testing.

(U) After FAT, NIQU participates as a member of the overall test team to plan and execute pre- operational testing of the integrated architecture. The NIQU topic lead will create a test plan for each capability that requires PQ testing. Each test plan will identify the specific test methodology, objectives, resource plan, risks and contingencies, and DRs to be verified. NIQU makes use of data that have passed through the NTO TE test architecture as well as data that have passed through the operational architecture (when possible) to assess the operational readiness of the enterprise. NIQU also makes use of the NSG Image

Quality Simulator (NIQS) as a gold standard reference product generator. NIQU documents results in a test report describing the results of the PQ testing. NIQU test participants and management conduct a NIQU Technical Review meeting to determine the NIQU go/no-go recommendation to provide to the ORR. Following that event, NIQU conducts an internal lessons learned exercise to capture any needed procedural or logistical changes to the standard test process. NIQU's analysis determines the level of format compliance, support data compliance, and the accuracy of products evaluated.

(U) Finally, during the ORR, NIQU provides an assessment of the NSG/ASG enterprise's PQ readiness for decision makers to consider in making transition decisions.

APPENDIX A. (U) NGA TEST ACTIVITIES

Table 2. (U) NGA Test Activities

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Test Activity	Purpose	Responsible	DoD Test Category
Unit and System (Factory) Integration Test	(U) Verify functionality of basic software units and ability of those software units to effectively operate together as a system	Developer	DT&E
Early Interface Test (EIT)	(U) Development contractor-to-contractor informal testing to assess a portion of the developers' products' requirements or capabilities as they interoperate with one or more other systems	Developer	DT&E
Inter-Segment Message Exchange (ISME)	(U//FOUO) The ISME subservice provides a hosted test environment to enable the NRO to assess interfaces via message exchange that could not be tested in the factory test environment.	Project Lead/Program Office	DT&E
Factory Acceptance Test (FAT)/Demonstration	(U) Demonstrate to the government that the entity performs as expected and all requirements have been met	Program Office and Developer	DT&E
Installation, Checkout, and Test (IC&T)	(U) Installation and checkout by the developer at the government test location to confirm successful configuration and connectivity of the system	Developer	DT&E
Integration Test (IT)	(U) Provides confidence that component systems have been successfully integrated, that their interfaces function correctly, and that they are ready for formal test in the enterprise architecture. Verification of fulfillment of User Story Acceptance Criteria.	Program Office and Developer	DT&E
Assisted Service IV&V	(U) The Assisted Service IV&V subservice provides a hosted test environment that provides a controlled operationally representative environment to enable customers to conduct testing of major upgrades (e.g., new functionality). This subservice is used for systems that have been assessed as mature in their development and testing processes.	Program Office/ Product Owner	OT&E
Assisted Service Regression Test	(U) The Assisted Service Regression Test subservice provides a hosted test environment that provides a controlled operationally representative environment to enable customers to conduct testing of minor upgrades (i.e., DR fixes).	Program Office/ Product Owner	OT&E
Performance Profiling	(U) The Performance Profiling subservice provides analysis of application performance to identify potential bottlenecks and/or aid application optimization. This subservice may include	Program Office and Developer	DT&E and OT&E

	ongoing performance audits to evaluate application performance against established benchmarks.		
Independent Verification and Validation (IV&V)	(U) Testing performed to validate the operability and interoperability of interfaces in newly delivered or changed systems, to ensure they will perform as expected and do no harm to the user's operational environment, and to verify fulfillment of Feature Acceptance Criteria	Government (NTO)	OT&E
Sensor Initialization	(U) The Sensor Initialization (Joint) subservice provides a review of data from new and/or existing sensor families for sensor performance. Assessment focuses on geolocation and mensuration accuracy, support data and visual rendition accuracy, and interoperability and functional performance throughout the Enterprise using an end-to-end data flow.	Government (NIQU)	OT&E
Image Quality (IQ) Test	(U) Testing performed to verify that the quality and format of imagery that is passed through the NSG/ASG is not negatively affected.	Government (NIQU)	DT&E and/or OT&E
Interoperability Test	(U) Testing performed to verify the ability of a new or modified system to effectively interoperate in the user's operational environment by satisfactorily exchanging information or services with other systems in the environment. Performed by NTO for systems being deployed internally in NGA and for basic interoperability of systems to be deployed external to NGA. Final interoperability testing for systems requiring DoDIIS certification is performed by JITC. Usually performed as a part of IV&V or regression testing.	Government (NTO/JITC)	OT&E
Security Test	(U) Test activities performed as a part of IV&V/Beta I and regression testing to verify that Director of Central Intelligence Directive (DCID) 6/3 security requirements have been met and that system vulnerabilities have been addressed as required.	Government (NTO/CAD)	OT&E
IAVA Test	(U) The Information Assurance Vulnerability Alert (IAVA) Test subservice provides an impact assessment of the security-driven software and configuration changes to the Web browsers and JAVA Runtime Environment (JRE) used by NGA.	Government (NTO)	OT&E
Joint Testing	(U) Enterprise Test and/or Joint Demonstration activities, conducted jointly with mission partner agencies, requiring NGA and one or more interfacing	Government (NTO)	OT&E

	organization to execute end-to-end threads to verify that end-user and community needs are met.		
Site Acceptance Test (SAT)	(U) A customer-planned and -conducted test that provides the user the opportunity to gain “hands-on” experience with a new capability before it is accepted operationally and that determines how well it meets operational needs and expectations.	Government (Customer)	OT&E
Operations Assessment	(U) Provides informal testing of interfaces, applications, and data using ad hoc manual procedures and/or automated testing on an operational network.	Government (NTO and Customer)	OT&E
Regression Test	(U) Although the system is deployed operationally, software maintenance releases, changes, updates, or patches will be originated by the development contractor, Operations & Sustainment (O&S) contractor, or ISP to resolve system DRs or other issues; these changes are tested to ensure they correct the deficiencies as intended, and do not negatively affect the enterprise.	Government (NTO)	OT&E
Emergency Test	(U) The Emergency Test subservice provides support for short-notice tests that are executed to verify correction of high-priority (i.e., Priority 1 or 2) DRs (including degradation of system performance) that were discovered in operations. Emergency tests are accorded special priority for scheduling to permit the fix to be delivered into operations with minimal delay.	Government (NTO and NIQU)	OT&E
Agile Test	(U) The Agile Test subservice provides full-service testing and/or analysis of entities developed using an agile methodology.	Government (NTO)	OT&E
DevOps Test	(U) DevOps is the practice of development and operational engineers working together during the entire service lifecycle, from design through the development process to production support. It is dependent upon efficient and effective conveyance of information within an automated process of development using automated processes and tools.	Government (NTO and NIQU)	OT&E
Web Services Test	(U) The Web Services Test subservice provides testing of new and modified web services used by NSG/ASG systems. Using test automation tools, service functionality is exercised by executing scripted requests to produce known service responses.	Government (NTO)	OT&E

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APPENDIX B. (U) ABBREVIATIONS AND ACRONYMS

A

A	(NGA) Analysis Directorate
ACAS	Assured Compliance Assessment Solution
ACAT	Acquisition Category
AIS	Automated Information
AoA	Analysis of Alternatives
APB	Acquisition Program Baseline
ASC	Analytical Services Contract
ASD	Assistant Secretary of Defense
ASG	Allied System for Geospatial-Intelligence
ASGB	Apps, Services Governance Board
ASP	Application Service Provider
AWS	Amazon Web Services

B

BAL	Business Area Lead
BPM	Business Process Management

C

C2S	Commercial Cloud Services
C3I	Command, Control, Communications, and Intelligence
CA	Computer Associates
CAD	(NGA) Cyber Assessments Division
CCB	Configuration Control Board
CDD	Capability Development Document
CDR	Critical Design Review
CI	Configuration Item
CINC	Commander in Chief
CIO	Chief Information Officer
CIO-T	Chief Information Officer, Information Technology Services
CJCSI	Chairman of the Joint Chiefs of Staff Instruction
CM	Configuration Management
COCOM	Combatant Command

COE	Common Operating Environment
COI	Critical Operational Issue
CONOPS	Concept of Operations
COTS	Commercial Off-the-Shelf
CR	Change Request
CS ERB	Consolidated Systems Engineering Review Board
CtF	Certification to Field
CTP	Critical Technical Parameter
CVE	Common Vulnerabilities and Exposures
D	
DBT	Define, Build, Test
DCID	Director, Central Intelligence Directive
DevOps	Development and Operations
DISA	Defense Information Systems Agency
DITL	Day-in-the-Life
DITSCAP Process	DoD Information Technology Security Certification and Accreditation
DNI	Director National Intelligence
DoD	Department of Defense
DoDD	Department of Defense Directive
DoDI	Department of Defense Instruction
DoDIIS	Department of Defense Intelligence Information System
DQR	Document Quality Review
DR	Discrepancy Report
DT&E	Developmental Test and Evaluation
D/TAC	Director, Integration Test & Image Quality Office
E	
ECP	Engineering Change Proposal
EIT	Early Interface Test
ERB	Engineering Review Board
ET	Enterprise Test
EWG	Engineering Working Group

F

FAT Factory Acceptance Test

FOT&E Follow-on Operational Test and Evaluation

FSR Factory Service Representative

G

GDS GEOINT Data Transformation Service

GEOINT Geospatial Intelligence

GFMSA GEOINT Functional Manager Standards Assessment

GIAT Geospatial Intelligence Advancement Testbed

GIMS GEOINT Information Management Services

GOTS Government Off-the-Shelf

GovCloud Government Cloud

GUI Graphical User Interface

GVS GEOINT Visualization Services

H

HP Hewlett Packard

HP ALM Hewlett Packard Application Lifecycle Management

HP LR Hewlett Packard Load Runner

HP UFT Hewlett Packard Unified Functional Test

I

IA Imagery Analyst

IAVA Information Assurance Vulnerability Alert

IAVB Information Assurance Vulnerability Bulletin

IAVT Information Assurance Vulnerability Technical Advisories

IC Intelligence Community

IC ITE Intelligence Community Integrated Test Environment

IC&T Installation, Checkout, and Test

IOC Initial Operational Capability

IOT&E Initial Operational Test and Evaluation

IoT Internet of Things

IPT Integrated Product Team

IQ	Image Quality
IQE	Image Quality Equation
ISME	Inter-Segment Message Exchange
ISP	Infrastructure Service Provider
IT	Integration Test
ITRR	(NTO) Internal Test Readiness Review
IV&V	Independent Verification and Validation
J	
JITC	Joint Interoperability Test Command
JRE	JAVA Runtime Environment
JROC	Joint Requirement Oversight Committee
JWICS	Joint Worldwide Intelligence Communications System
K	
KPP	Key Performance Parameter
M	
MDA	Milestone Decision Authority
MOE	Measure of Effectiveness
MOP	Measure of Performance
MOS	Measure of Suitability
N	
NCE	NGA Campus East
NCI&A	NGA Cloud Integration & Adoption
NCL	NSG Consolidated Library
NCMP	NGA Configuration Management Plan
NCW	NGA Campus West
NDC	NSG Data Center
NEC	NIQU Engineering Cell
NEF	NIQU Evaluation Facility
NESG	NSG Enterprise Scenario Generator
NGA	National Geospatial-Intelligence Agency
NGA CCB	NGA Configuration Control Board
NGANet	NGA Network

NIIRS	National Imagery Interpretability Rating Scale
NIIST	NTO Intelligence Information Support Team
NIQS	NSG Image Quality Simulator
NIQU	NGA Image Quality and Utility
NIST-SP	National Institute of Standards and Technology Special Publication
NITF	National Imagery Transmission Format
NORD	NGA's Operational Requirements Document
NRAI	NSG Requirements for Application and Infrastructure
NRO	National Reconnaissance Office
NSG	National System for Geospatial Intelligence
NTM	National Technical Means
NTO	NGA Test Organization
NTO TE	NTO Test Environment
NTPM	NTO Test Planning Meeting

O

OGC	Office of the General Counsel
OL	Operations Lead
OMR	Organizational Management Review
ORD	Operational Requirements Document
O&M	Operations and Maintenance
O&S	Operations and Sustainment
OT&E	Operational Test and Evaluation

P

PIM	Problem Incident Management
PM	Program/Project Manager
PMO	Program/Project Management Office
PO	Product Owner
POC	Point of Contact
POR	Program of Record
PQ	Product Quality
PQ-ORC	Product Quality Operational Readiness Criteria
PR	Problem Report

PSR	Pre-Ship Review
Q	
QA	Quality Assurance
QTP	Quick Test Professional
R	
R	(NGA) Research Directorate
R&D	Research and Development
RAM	Resource Allocation Meeting
RBT	Risk-Based Testing
RDT&E	Research, Development, Test, and Evaluation
RFC	Request for Change
RFS	Request for Service
RMF	Risk Management Framework
RMSE	Root-Mean-Square Error
RTM	Return to Mission
RVTM	Requirements Verification Traceability Matrix
S	
S	(NGA) Source Directorate
S&OT	Stability and Operability Testing
SAT	Site Acceptance Test
SBU	Sensitive But Unclassified
SC	Secret Cloud
SCAP	Security Content Automation Protocol
SD	System Demonstration
SDPS	Softcopy Display Performance Standard
SECNet	Secret Network
SEFS	Softcopy Exploitation Facilities Standard
SEIT	System Engineering Integration Team
SI	(NGA) Security and Installation Operations Directorate
SIT	Systems Integration Testing
SME	Subject Matter Expert

SRD	Segment/Specification Requirements Document
SSAA	System Security Authorization Agreement
SSP	System Security Plan
SWAP	Software Assurance Process
T	
TA	Architecture and Engineering Group
TAC	Integration Test & Image Quality Office
TACA	Assessment Division
TACQ	Image Quality Division
TAC-SA	Integration Test & Image Quality Office Systems Administrator
TBD	To Be Determined
TBR	To Be Resolved
TC	Top Secret Cloud
TD	Test Director
TEMP	Test and Evaluation Master Plan
TM	Technical Manual
TOR	Terms of Reference
TRR	Test Readiness Review
TSR	Cyber Security
T&E	Test and Evaluation
T/OCT	Transition/Operations Confidence Test
U	
UC	Unclassified Cloud
US	United States
USCYBERCOM	United States Cyber Command
V	
VOLT	Verification and Validation On-Line Tools

APPENDIX C. (U) GLOSSARY**(U) Absolute National Imagery Interpretation Rating Scale (NIIRS) Method**

The most resource intensive, method for determining the level of interpretability of a product delivered by an imaging enterprise. With the Absolute NIIRS rating or Criteria rating method, a relatively large group of imagery analysts compare imagery of unknown NIIRS to the set of word criteria which define each numerical level of the NIIRS scale. Each of the analysts assigns the test image a decimal NIIRS rating which, in their professional opinion, satisfies all the criteria for a given NIIRS level. The accumulation of NIIRS ratings for a given image is statistically analyzed to determine the NIIRS of the test image. Achieving highly accurate NIIRS ratings for a single image requires a large number of Absolute NIIRS ratings in order to overcome the inherent variability between imagery analyst ratings.

(U) Architecture 1. The structure of components, their interrelationships, and the principal guidelines governing their design and evolution over time. 2. The highest level of the NSG/ASG program, including all NGA segments. The term used in T&E to include all NGA entities and "external" entities that interface with NSG/ASG.

(U) Availability A measure of the degree to which an item is in an operable state and can be committed at the start of a mission when the mission is called for at an unknown (random) point in time.

(U) Baseline 1. A specification or product that has been formally reviewed and agreed upon, thereafter serves as the basis for further development, and can be changed only through formal change control procedures. 2. The currently approved operational or effective version of a system, application, network or specification document. 3. Defined quantity or quality used as the starting point for subsequent efforts and progress measurement that can be a technical, cost, or schedule baseline.

(U) Capability A measure of the system's ability to achieve mission objectives, given the system condition during the mission.

(U) Capability Development Document (CDD) A formatted document prepared by the user that builds upon the Initial Capabilities Document (ICD) to identify operational performance parameters necessary to design an affordable and militarily useful increment of capability. In addition to identifying performance parameters for the current increment (including KPPs), the CDD will outline the overall strategy to develop the full or complete capability through future increments. The CDD supports the Milestone B decision (DoD).

(U) Commercial Off-The-Shelf (COTS) Commercial items that require no unique government modifications or maintenance over the life cycle of the product to meet the needs of the procuring agency.

(U) Compatibility The capability of two or more items or components of equipment or material to exist or function in the same system or environment without mutual interference.

(U) Configuration A collection of an item's descriptive and governing characteristics that can be expressed in functional terms, i.e., what performance the item is expected to achieve; and in physical terms, i.e., what the item should look like and consist of when it is built.

(U) Configuration Management (CM) The technical and administrative direction and surveillance actions taken to identify and document the functional and physical characteristics of a configuration item (CI), to control changes to a CI and its characteristics, and to record and report change processing and implementation status. It provides a complete audit trail of decisions and design modifications.

(U) Critical Design Review (CDR) A review that may be conducted to determine that the detailed design satisfies the performance and engineering requirements of the development specification; to establish the detailed design compatibility among the item and other items of equipment, facilities, computer programs and algorithms, and personnel; to assess producibility and risk areas; and to review the preliminary product baseline specifications. Normally conducted during the System Development and Demonstration phase.

(U) Critical Operational Issue (COI) A key operational effectiveness and/or operational suitability issue (not a parameter, objective, or threshold) that must be examined in operational test and evaluation (OT&E) to determine the system's capability to perform its mission. A COI is normally phrased as a question that must be answered in order to properly evaluate operational effectiveness (e.g., "Will the system detect the threat in a combat environment at adequate range to allow successful engagement?") or operational suitability (e.g., "Will the system be safe to operate in a combat environment?").

(U) Critical System Characteristics Unique support concepts resulting in special test and analysis requirement (e.g., post-deployment software support, hardness against nuclear effects; resistance to countermeasures; resistance to reverse engineering/exploitation efforts (anti-tamper); and development of new threat simulation, simulators, or targets).

(U) Critical Technical Parameter (CTP) Critical Technical Parameters are measurable critical system characteristics that, when achieved, allow the attainment of operational performance requirements. They are technical measures derived from Operational Requirements Document (ORD) requirements.

(U) Delta NIIRS Rating Method An alternative method for determining the level of interpretability of a product delivered by an imaging enterprise based on determining the relative NIIRS difference between two similar images. In Delta NIIRS, a NIIRS certified imagery analyst views two images either side by side on one image display, or flickers between two closely registered images, and assesses the NIIRS difference between the two. Example: Two images which are only just perceptibly different are often rated as having a 0.1 delta-NIIRS.

(U) Development The process of working out and extending the theoretical, practical, and useful applications of a basic design, idea, or scientific discovery.

Design, building, modification, or improvement of the prototype of a vehicle, engine, instrument, or the like as determined by the basic idea or concept. Includes all efforts directed toward programs being engineered for Service use but which have not yet been approved for procurement or operation, and all efforts directed toward development engineering and test of systems, support programs, vehicles, and weapons that have been approved for production and service deployment.

(U) Developmental Test and Evaluation (DT&E) 1. Any testing used to assist in the development and maturation of products, product elements, or manufacturing or support processes. 2. Any engineering-type test used to verify status of technical progress, verify that design risks are minimized, substantiate achievement of contract technical performance, and certify readiness for initial operational testing. Development tests generally require instrumentation and measurements, and are accomplished by engineers, technicians, or soldier operator-maintainer test personnel in a controlled environment to facilitate failure analysis.

(U) DoD Instruction (DoDI) 5000.2 "Operation of the Defense Acquisition System." Establishes the framework, procedures, and general approach for the management of all acquisition programs and technology projects. Specifically authorizes milestone decision authorities (MDAs) to tailor programs to meet cost, schedule, and performance goals consistent with statutory requirements.

(U) Discrepancy Report (DR) A documented description of a deficiency identified in the development, test, and/or operational use of a configuration item throughout its life cycle. Typically associated with NGA tracking mechanism for issues/concerns. DRs can also be associated with problem reports (PRs), software trouble reports (STRs), and hardware failure reports (HFRs).

(U) Early Interface Testing (EIT) Testing conducted during the software development cycle, using developmental/engineering software, to assess interfaces external to the system or product. This testing is contractor to contractor, with no pass/fail criteria.

(U) Effectivity A software/hardware delivery of new functions or capabilities, which are associated together and work together, even if different NGA offices and their developers are responsible for the components. The delivery is made on an established schedule.

(U) Effectiveness A measure of how well the system achieves its purpose. Dependent upon the performance of the component parts, indicates how well they are integrated, how well the system and the people work together, and how well the system is supported.

(U) Engineering NIIRS Standard (ENS) Rating Method Considered the most accurate method for determining the level of interpretability of a product delivered by an imaging enterprise. This method relies on a predefined set of image chips, each of which has been assigned a NIIRS rating by using the absolute NIIRS

method, so that their NIIRS is very well understood. The ENS set consists of approximately a dozen image chips for each order of battle which are evenly spaced in quality (about $\frac{3}{4}$ NIIRS separation) and which span the range of quality expected to be received by NGA imagery analysts. Each image chip is assigned a letter code to facilitate the rating process. To obtain an ENS rating for an image of unknown NIIRS an imagery analyst compares the unknown image to the image chips of the appropriate set of ENS chips and selects the two adjacent chips which bracket the quality of the test image. The analyst then assesses, on a 10-point scale, the relative quality of the unknown chip compared to the two ENS marker chips. The ENS ratings are then converted to NIIRS values using the known NIIRS of the two marker chips and the analyst's relative quality assessment. The advantage of the ENS rating method is that it has lower variability and therefore requires fewer imagery analyst ratings in order to achieve a desired rating accuracy.

(U) Entity Throughout this document, reference is frequently made to software, hardware, systems, applications, services, desktops, corporate systems, and/or networks. For simplification and clarity, the term "entity" is used to represent any of those categories as related to NGA testing, unless reference is being made to a specific entity or category.

(U) Exploitation The process of obtaining usable data from imagery or other Geospatial data sources.

(U) Factory Acceptance Test (FAT) A contractor-conducted test to demonstrate to the program office that all contracted hardware and software requirements have been met. FAT is conducted at the development facility and includes all aspects of systems testing as well as interoperability (if appropriate), security, and performance testing.

(U) Follow-On Operational Test and Evaluation (FOT&E) The test and evaluation (T&E) that may be necessary after the Full Rate Production Decision Review to refine the estimates made during initial operational test and evaluation (IOT&E), to evaluate changes, and to re-evaluate the system to ensure that it continues to meet operational needs and retains its effectiveness in a new environment or against a new threat.

(U) Functional Baseline Documentation describing system/product functional characteristics and the verification required to demonstrate the achievement of those specified functional characteristics. The system or product specification establishes the functional baseline.

(U) Hardware When related to computers: The physical equipment that makes up a computer system (e.g., terminals and storage devices, as opposed to programming software).

(U) Independent Verification and Validation (IV&V) An independent review of software performed by an organization that is technically, managerially, and financially independent of the development organization.

(U) Information Assurance Information operations that protect and defend information and information systems by ensuring their availability, integrity, authentication, confidentiality, and non-repudiation. This includes providing for the restoration of information systems by incorporating protection, detection, and reaction capabilities.

(U) Information Operations Actions taken to affect adversary information and information systems while defending one's own information and information systems.

(U) Information System A discrete set of information resources (e.g., personnel, data, software, computers, communications equipment) organized for the collection, processing, maintenance, use, sharing, dissemination, or disposition of information.

(U) Initial Operational Capability (IOC) The first attainment of the capability to employ effectively an item of equipment or system of approved specific characteristics with the appropriate number, type, and mix of trained and equipped personnel necessary to operate, maintain, and support the system. It is normally defined in the Operational Requirements Document (ORD).

(U) Initial Operational Test and Evaluation (IOT&E) Dedicated operational test and evaluation conducted on production, or production-representative articles, to determine whether systems are operationally effective and suitable, and which supports the decision to proceed beyond low rate initial production.

(U) Integration Test Refers to the process of ensuring products work correctly when brought into an integrated, operational environment; that they do not adversely affect one another; and that they conform to the standards and specifications described in this document.

(U) Integration Test Baseline The version of the system delivered to the NTO after FAT and used for DT&E Phase 2 testing. Once integration testing is completed, the updated integration test baseline becomes the operational test baseline.

(U) Interface The functional and physical characteristics required to exist at a common boundary or connection between persons, or systems, or between persons and systems.

(U) Interoperability 1. The ability of systems, units, or forces to provide services to, or accept services from, other systems, units, or forces and to use the services so exchanged to enable them to operate effectively together. Interoperability is a mandatory KPP. (DoDI 5000.2/CJCSI 3170.01A/CJCSI 6212.01F) 2. The conditions achieved among communications-electronics systems, or items of communications-electronics equipment, when information or services can be exchanged directly and satisfactorily between them or their users. The degree of interoperability should be defined when referring to specific cases. (CJCSI 3170.01A/CJCSI 6212.01F)

(U) Joint Testing Test activities that require NGA and another interfacing organization to execute end- to-end threads that prove end user and community needs are met.

(U) Key Performance Parameters (KPP) KPPs are a critical subset of the performance parameters found in the ORD, and are included in the performance portion of the Acquisition Program Baseline (APB). Each KPP has a threshold and an objective value. KPPs represent those capabilities or characteristics so significant that failure to meet the threshold value of performance can be cause for the concept or system selected to be re-evaluated or the program to be reassessed or terminated. KPPs are validated by the Joint Requirement Oversight Committee (JROC) for Acquisition Category (ACAT) I and ACAT IA programs.

(U) Life cycle All phases of the system's life including research, development, test and evaluation (RDT&E), production, deployment (inventory), operations and support (O&S), and disposal.

(U) Maintainability The ability of an item to be retained in, or restored to, a specified condition when maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair.

(U) Measure of Effectiveness (MOE) 1. A measure of operational success that must be closely related to the objective of the mission or operation being evaluated. MOEs denoted in the Analysis of Alternatives (AoA), Operational Requirements Document (ORD), and Test and Evaluation Master Plan (TEMP) must be consistent. A meaningful MOE must be quantifiable and a measure to what degree the real objective is achieved. 2. Refers to the process of ensuring products work correctly when brought into an integrated, operational environment; that they do not adversely affect one another; and that they conform to the standards and specifications described in this document.

(U) Measures of Performance (MOP) 1. Measures of a system's technical performance expressed as speed, payload, range, time on station, frequency, or other distinctly quantifiable performance features. Several MOPs may be related to the achievement of a particular MOE. 2. A quantitative measure of the lowest level of physical performance.

(U) Measures of Suitability (MOS) The degree to which a system can be placed satisfactorily in planned operational environment field use with consideration given to reliability, maintainability, availability, transportability, interoperability, compatibility, human factors, documentation, training, logistics support, and safety.

(U) Milestone Decision Authority (MDA) The individual designated in accordance with criteria established by the Under Secretary of Defense (Acquisition, Technology and Logistics) (USD (AT&L)) or by the Assistant Secretary of Defense (Command, Control, Communications, and Intelligence) (ASD (C3I)) for automated information system (AIS) acquisition programs, to approve entry of an acquisition program into the next phase. (DoDI 5000.2)

(U) Mission The objective or task, together with the purpose, which clearly indicates the action to be taken.

(U) National System for Geospatial-Intelligence (NSG) The extensive network of systems used by the DoD and Intelligence communities that share and exploit imagery, imagery intelligence, and Geospatial information. These systems provide capabilities for integration, management, collection, exploitation and production, dissemination, and archive and infrastructure for this information. Organizations that have some level of interaction with NSG, but are not part of DoD and IGC, are considered participants in NSG if they adhere to the technical and system standards.

(U) NSG Image Quality Simulator (NIQS) A software system which emulates the key image handling functions of the end-to-end NSG/ASG enterprise. NIQU uses the NIQS to generate Gold Standard versions of imagery for use in assessing the performance of NSG/ASG segment image handling implementations. The NIQS also provides capabilities to support data checkouts by dumping the contents of a product's support data out in a manner which allows detailed examination.

(U) Numerical Comparison Method A quantitative method for assessing the degradation of an image that has undergone some sort of image handling process. In this test a copy of the image prior to the image handling process is subtracted from the image after the image handling treatment and the differences are squared on a pixel by pixel basis and summed. This process calculates the root-mean-square error (RMSE) between the two images. For image handling processes which should have no numerical difference the acceptable RMSE is zero. If two images handled by different hardware are compared, the acceptable RMSE is 5 or lower.

(U) Operational Baseline/Configuration Item List A list that consists of documents/data that describes the operation and maintenance of the delivered and installed segment to include installation, operations, maintenance and user manuals. It also includes a list of the "as built" hardware, software, firmware, and COTS/GOTS products that comprise the operational segment or component. Proposed changes to this list and additions to or deletions from, require approval by the responsible Segment manager, as delegated by the NGA CCB. This baseline configuration item list is also sub-divided alphabetically in accordance with the NGA CCB organizational structure.

(U) Operational Capability The measure of the results of the mission, given the condition of the systems during the mission (dependability).

(U) Operational Effectiveness The overall degree of mission accomplishment of a system when used by representative personnel in the environment planned or expected (e.g., natural, electronic, or threat) for operational employment of the system, considering organization, doctrine, tactics, survivability, vulnerability, and threat (including countermeasures, initial nuclear weapons effects, nuclear, biological, and chemical contamination (NBCC) threats).

(U) Operational Enterprise Network The NGA SBU and NGANet networks providing office automation and production capabilities. Specifically excluded are test and laboratory networks (e.g., Research's Geospatial Intelligence Advancement Testbed (GIAT)).

(U) Operational Requirements User- or user representative-generated validated needs developed to address mission area deficiencies, evolving threats, emerging technologies or weapon system cost improvements. Operational requirements form the foundation for weapon system unique specifications and contract requirements.

(U) Operational Requirements Document (ORD) A formatted statement containing performance and related operational performance parameters for the proposed concept or system.

(U) Operational Suitability (OS) The degree to which a system can be placed satisfactorily in field use with consideration being given to availability, compatibility, transportability, interoperability, reliability, wartime usage rates, maintainability, safety, human factors, manpower supportability, logistic supportability, natural environmental effects and impacts, documentation, and training requirements.

(U) Operational Test and Evaluation (OT&E) The field test, under realistic conditions, of any item (or key component) of weapons, equipment, or munitions for the purpose of determining the effectiveness and suitability of the weapons, equipment, or munitions for use in combat by typical military users and the evaluation of the results of such tests.

(U) Operational Test Baseline The locked-down version of the system software that undergoes government testing/OT&E. Upon successful completion of government testing, the updated operational test baseline becomes the product baseline.

(U) Performance Those operational and support characteristics of the system that allow it to effectively and efficiently perform its assigned mission over time. The support characteristics of the system include both supportability aspects of the design and the support elements necessary for system operation.

(U) Problem Report (PR) A documented description of a defect, unusual occurrence, observation, or failure that requires investigation and may require software modification, typically associated with factory tracking of issues/concerns. (See Discrepancy Report (DR))

(U) Product Baseline The version of an entity's software that has successfully completed government testing/OT&E and has received a certification to field. Copies of the product baseline are also referred to as Gold Copies and are maintained by both the NTO and the developer.

(U) Production Data Data, which is produced through the complete cycle, that will be used in the operational system.

(U) Program Manager (PM) The individual designated in accordance with criteria

established by the appropriate Component Acquisition Executive to manage an acquisition program, and appropriately certified under the provisions of the Defense Workforce Improvement Act. A PM has no other command or staff responsibilities within the Component. (Department of Defense Directive (DoDD) 5000.1)

(U) Quality The composite of materiel attributes, including performance features and characteristics of a production or service to satisfy a customer's given need.

(U) Reliability The ability of a system and its parts to perform its mission without failure, degradation, or demand on the support system.

(U) Site Acceptance Test (SAT) A user conducted test which provides the user the opportunity to gain "hands-on" experience with a new capability before it is transitioned to operations, and to determine (usually qualitatively) how well it meets operational needs and expectations. SAT is conducted on-site at the user's location and includes all aspects of systems testing including interoperability, security, and performance testing. A user-designated SAT lead is responsible for developing test procedures and/or day-in-the life work-flows and scenarios with which to exercise the system. User personnel will usually conduct the SAT with SME guidance from development contractor personnel.

(U) Testing an element of inspection. Generally, denotes the determination by technical means of the properties or elements of supplies, or components thereof, including functional operation, and involves the application of established scientific principles and procedures.

(U) Test Readiness Review (TRR) A review to evaluate and verify that a program is prepared to proceed with formal testing for one or more configuration items. Typically held prior to software qualification testing for critical Software Configuration Items.

(U) Test Report Formally documents the results, conclusions, and recommendations as a result of each phase of development testing (DT)/operational testing (OT).

(U) Threshold The minimum acceptable value that, in the user's judgment, is necessary to satisfy the need. If threshold values are not achieved, program performance is seriously degraded, the program may be too costly, or the program may no longer be timely. The spread between objective and threshold values is individually set for each program based on the characteristics of the program (e.g., maturity, risk). If the threshold values are not otherwise specified, the threshold value for performance will be the same as the objective value, the threshold value for schedule will be the objective value plus six months for ACAT I programs and three months for ACAT IA programs, and the threshold value for cost will be the objective value plus 10 percent.

(U) To Be Determined (TBD) A valid requirement that has not been fully defined. It may be proposed against to form part of a contractual baseline under explicitly stated assumptions. When a TBD is removed as a result of an RFC, the impact of

removal shall be assessed by all affected contractors in the same manner as any other change in requirements.

(U) To Be Resolved (TBR) A valid requirement that is imposed but is subject to change. Notation used to indicate that the functionality or numerical performance level specified in this requirement is subject to marginal change but that the overall requirement is unlikely to disappear entirely from this portion of the document. The functionality or performance value used in the requirement is part of the baseline. Any TBR work-off plan should indicate the trade studies and any relationships between TBRs known to the document users. When this TBR is removed as result of an RFC, all affected parties shall assess the impact.

(U) Training The level of learning required to adequately perform the responsibilities designated to the function and accomplish the mission assigned to the system.

(U) User An operational command or agency that receives or will receive benefit from the acquired system. Commanders-in-Chief (CINC) and their Service component commands are the users. There may be more than one user for a system. The Service component commands are seen as users for systems and organize, equip, and train forces for the CINCs of the unified commands. The Chiefs of Services and heads of other DoD components are validation and approval authorities and are not viewed as users. (CJCSI 3170.01A).

(U) Validation 1. The review of documentation by an operational authority, other than the user, to confirm the need or operational requirement. (CJCSI 3170.01A)
2. The process by which the contractor (or as otherwise directed by the DoD component procuring activity) tests a publication/technical manual (TM) for technical accuracy and adequacy. 3. The process of evaluating a system or software component during, or at the end of, the development process to determine whether it satisfies specified requirements.

(U) Verification The process of evaluating a system or software component to determine whether the products of a given development phase satisfy the (exit) conditions imposed at the start of that phase.

(U) Vulnerability The characteristics of a system that cause it to suffer a definite degradation (loss or reduction of capability to perform the designated mission) as a result of having been subjected to a certain (defined) level of effects in an unnatural (man-made) hostile environment. Vulnerability is considered a subset of survivability.

APPENDIX D. (U) REFERENCES

D.1 (U) INTERNAL NGA DOCUMENTS

- (U) NCMP-E, National Geospatial-Intelligence Agency (NGA) Configuration Management Plan (NCMP)
- (U) Tier 2 National Geospatial-Intelligence Agency (NGA) Enterprise Readiness (ER) Process & Procedures
- (U) NGA's Enterprise Systems Engineering Plan (SEP) for the NSG
- (U) NGA's NSG Online Workspace (NOW) Development and Test Environment Concept of Operations / Management Plan
- (U) NGA's Operational Requirements Document (NORD) Updated Capabilities – November 2004 (U) NGA.STND.0014_2.2 Softcopy Image Processing Standard
- (U) NGA.STND.0015_3.1 Display Performance Standard
- (U) NGA.STND.0016_2.0 Softcopy Exploitation Facility Standard
- (U) National Geospatial-Intelligence Agency (NGA) Test and Evaluation (T&E) - Tier 2 Process
- (U) Softcopy Quality Assurance Process, NGA Image Quality and Utility Program
- (U) Consolidated NGA (CoNGA) Security Classification Guide
- (U) The Test Director Handbook

D.2 (U) EXTERNAL DOCUMENTS

- (U) Defense Intelligence Integrator's Guide (DIIG)
- (U) Department of Defense Intelligence Information System (DoDIIS) Instruction 2000
- (U) DoD 5200.40, Department of Defense Information Technology Security Certification and Accreditation Process (DITSCAP)
- (U) Intelligence Community Directive (ICD) 503 Risk Management Framework
- (U) DoDI 5000.02, Operation of the Defense Acquisition System
- (U) DoD Interim Defense Acquisition Guidebook (U) GEOINT Integrated Test and Transition Plan (U) Risk Management Guide for DoD Acquisition
- (U) Test and Evaluation Policy for DoD Intelligence Information System (DoDIIS) Intelligence Mission Applications (IMAs)